Nimbus360DashGen

Project Team

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Chapter 1

Introduction

Identify the product or application whose requirements are specified in this document. Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers. [**?**].

1.1 Problem Statement

In recent times, Pakistani restaurants have seen increasing competition and operational challenges. These challenges include the management of inventory, processing sales in an efficient manner, forecasting demand and responding to regional preferences, events and seasonal variations. Further, the competition in the food industry has seen a surge post covid 19 [**?**]. Traditionally, the decision-making methods relied on manual data analysis and external consultants. Both of the listed solutions result in delayed responses, increased spending on human resources, and limited scalability. This arises a need for an integrated solution which points an all-in-one platform to the business owners which enables them to make data-driven decisions. This helps a business to grow efficiently and independently while cutting down the costs of managing extra human resources.

1.2 Scope

Our project is developing a complete software solution and features advanced analytics, machine learning combined with POS and warehouse management functionalities. Here, the answer will need to require data directly from Foodpanda. We will use the foodpanda.pk website to get the competitive analysis, build tailor made business insights and trend predictions The project will be executed in four phases with a number of validations featured for every phase to know how well it works and scales.

1.3 Modules

1. Web Scraper
2. Point of Sale (POS) Module
3. Kitchen Portal Module
4. Manager Portal Module
5. Administration Portal Module
6. Warehouse Portal

1.3.1 Web Scraper

We will code a web scraper to get the data from foodpanda and deliver it in the form of an api endpoint to our model to retrain whenever required or the data is updated. Only the publicly available data will be scrapped for training purposes.

1. Access the foodpanda.pk website to access all the possible cities.
2. Store the names of all the possible locations and then store the names of all possible restaurants in the area.
3. Get the details of every single restaurant.

1.3.2 Point of Sale (POS) Module

This module will handle the processing of orders alongside all payment transactions integrated into them and customer management. It will feature an interface for restaurant staff to manage sales effectively.

1. Cart Management
2. Customer Management
3. Order Management
4. Receipt Generation

5. Report Generation
6. Complaint Portal

1.3.3 Kitchen Portal Module

This module includes the analysis of kitchen's performance in terms of order preparation time and customer feedback analysis. Primarily, we will focus on keeping track of time and the kitchen performances.

1. Mark orders as complete.
2. View active orders
3. Priority based changes in UX.

1.3.4 Manager Portal Module

The module highlights the manager operations in a restaurant's branch. Further, the specified module includes the analysis of all the recommendations provided by our system about the branch and these will be forwarded to the administration upon selection by the manager.

1. Branch management
2. Cashier Management
3. Product Management
4. Recommendations
5. Complaint Resolution Portal

1.3.5 Administration Portal Module

The Administrator Portal has all of the features that an administrator requires ranging from adding a new branch to communicating with the managers.

1. Predictive Analysis - Predicting the future of the restaurant or a specific branch will be catered in this module.

2. Recommender System - Providing recommendations for a new branch, the area to target, updating the menu, adjusting the prices, deals with respect to a new event or change in season and a more efficient finance solution will be catered in this module.

3. Competitive Analysis - Locality and competition matters. This module includes the analysis of the competitors in your niche and figuring out a solution for your restaurant to be at a much better position.

1.3.6 Warehouse Portal

The warehouse is responsible for the inbound and outbound of products for the branches.

1.4 User Classes and Characteristics

User Class	Description
Kitchen Staff	The Kitchen Staff receives orders placed by customers and prepares the meals. Orders are prioritized based on the time left for completion, as per the thresholds set by the Branch Manager. Once the meal is ready, the Kitchen Staff marks the order as prepared and ready for pickup. They also provide statistics on kitchen performance, which are analyzed for optimization.
Branch Manager	The Branch Manager oversees operations at a specific branch. They are responsible for setting preparation time thresholds for kitchen orders, managing inventory needs, generating sales, order, and custom reports, and ensuring efficient operation. Branch Managers also assign tasks to staff, monitor order prioritization, and forward necessary reports to the Admin for review.
Cashier	The Cashier processes customer orders and payments at the branch. They handle order placements, modifications, and cancellations. They also manage order receipts and ensure the accurate tracking of payments. Cashiers are responsible for updating the order status after payment and forwarding it to the Kitchen Staff for preparation.
Warehouse Manager	The Warehouse Manager oversees the management of inbound and outbound inventory. They track stock received at the warehouse and ensure proper distribution to various branches based on demand forecasts. The Warehouse Manager is responsible for maintaining real-time inventory updates, avoiding shortages or excesses, and ensuring efficient logistics between the warehouse and branches.
Admin	The Admin has authority over the entire system, including managing branches, assigning Branch Managers, generating high-level reports such as sales and performance metrics, and ensuring the system's smooth operation. They also approve reports received from Branch Managers and monitor overall performance.. They can do different types of analysis such as predictive, competitive, and feedback analysis and receive recommendations based on the analysis completed.

Table 1.1: User Classes and Descriptions

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

Section	**Content**
Designation	UC-CM-01
Name	ManageCart
Author	Fatima Sarmad
Priority	High
Criticality	Medium
Source	Customer request, menu selection, order review
Responsible	Cashier
Description	The restaurant staff or cashier manages the customer's cart by adding, removing, or reviewing items.
Trigger Event	• The customer requests to modify their order (add or remove items). • The customer requests to review the cart before checkout.
Actors	Cashier
Pre-Condition	• The POS system is operational. • The customer has selected at least one item from the menu. • The staff or cashier is logged into the system.
Post-Condition	• The cart is updated with the latest changes (items added or removed). • The updated cart is displayed with recalculated totals.
Result	The customer's order is updated, ready for further modifications or checkout.
Alternative Scenario	• Multiple items are added or removed from the cart at once. • The cashier modifies the quantity of an item in the cart. • Promotional offers are applied based on the cart total.
Exception Scenario	• The system fails to add or remove an item • The selected item is out of stock • The cart is empty

Table 2.1: ManageCart Use Case

Section	Content
Designation	UC-CM-03
Name	Process Payment
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Checkout, order placed
Responsible	Cashier
Description	After finalizing the cart, the cashier processes payment either by cash or card. Once payment is confirmed, the transaction is completed and a receipt is generated.
Trigger Event	The cashier has entered all items requested by the customer and total has been made.
Actors	Cashier
Pre-Condition	1. Items are finalized in the cart. 2. Payment method (cash/card) is available. 3. The POS system is operational and connected to the payment gateway.
Post-Condition	1. Payment is successfully processed. 2. A receipt is generated. 3. The order is marked as completed.
Result	The payment is recorded, and the receipt is generated.
Alternative Scenario	• The customer splits the payment between cash and card. • A discount or promotional offer is applied during checkout.
Exception Scenario	• Payment processing fails (e.g., insufficient funds, card error), and an error message is displayed. • The payment system is offline, and manual payment handling is required.

Table 2.2: Process Payment

Section	Contents
Designation	UC-CM-01
Name	Track Customer Orders
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Customer request, dining experience management
Responsible	Restaurant Staff, Manager
Description	Tracks customer orders across visits to allow staff to access past orders for personalized service and faster order taking.
Trigger Event	Customer places a new order or cashier checks customer's order history.
Actors	Cashier, Manager
Pre-Condition	• The customer is identified in the system. • Past orders are available in the system.
Post-Condition	• The customer's order history is displayed. • Staff can use past orders to assist in current orders or suggest personalized options.
Result	Enhanced customer experience with personalized service.
Alternative Scenario	• Customer reorders from past visits. • The system suggests "favorite orders" based on preferences.
Exception Scenario	• No order history available for a new customer. • The system fails to retrieve past orders due to technical errors.
Qualities	• Usability: Easily retrieves past orders.

Table 2.3: Track Customer Orders

Section	Content
Designation	UC-CM-02
Name	Track Customer Preferences
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Customer request, personalized service
Responsible	Manager
Description	Captures and tracks customer such as loyalty points, customer feedback, track orders.
Trigger Event	Customers place an order and the system updates preferences based on choices.
Actors	Manager
Pre-Condition	• Customer orders exist in the system. • Preferences have been logged from previous orders.
Post-Condition	• Customer preferences are updated and accessible. • System provides tailored recommendations.
Result	Personalized service with suggestions based on preferences.
Alternative Scenario	• System suggests menu items based on preferences. • Preferences are updated after each visit.
Exception Scenario	• No preferences available for new customers. • System fails to retrieve or update preferences.
Qualities	• Usability: Interface should allow easy update and viewing of preferences.

Table 2.4: Track Customer Preferences

Section	Content
Designation	UC-CM-03
Name	Customer Profile Management
Author	Fatima Sarmad
Priority	High
Criticality	Medium
Source	Customer data, loyalty programs
Responsible	Cashier, Manager, Administrator
Description	Allows responsible staff to create and manage customer profiles, storing contact info, order history, preferences, and loyalty points.
Trigger Event	New customer visits or cashier updates existing information of a customer.
Actors	Cashier, Manager, Admin
Pre-Condition	• Customer agrees to share details. • System can store customer information.
Post-Condition	• Customer profile is created or updated. • Loyalty points are tracked, and order history is stored for future use.
Result	Customer profiles stored, enabling personalized service and loyalty tracking.
Alternative Scenario	• Customer updates profile details, such as contact info or preferences. • Profile is automatically updated with new orders.
Exception Scenario	• Customers decline to share details, so no profile is created. • System fails to save or update the customer profile.
Qualities	• Usability: Simple interface for managing profiles.

Table 2.5: Customer Profile Management

Section	Content
Designation	UC-CM-03
Name	Customer Profile Management
Author	Fatima Sarmad
Priority	High
Criticality	Medium
Source	Customer data, loyalty programs
Responsible	Cashier, Manager, Administrator
Description	Allows responsible staff to create and manage customer profiles, storing contact info, order history, preferences, and loyalty points.
Trigger Event	New customer visits or cashier updates existing information of a customer.
Actors	Cashier, Manager, Admin
Pre-Condition	• Customer agrees to share details. • System can store customer information.
Post-Condition	• Customer profile is created or updated. • Loyalty points are tracked, and order history is stored for future use.
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Alternative Scenario	• Customer updates profile details, such as contact info or preferences. • Profile is automatically updated with new orders.
Exception Scenario	• Customers decline to share details, so no profile is created. • System fails to save or update the customer profile.
Qualities	• Usability: Simple interface for managing profiles.

Table 2.6: Customer Profile Management

Section	Content
Designation	UC-OM-02
Name	Order Fulfillment
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Kitchen Portal
Responsible	Kitchen Staff, Restaurant Staff
Description	After the order is placed, it is prepared in the kitchen and marked as completed when ready to serve.
Trigger Event	The kitchen receives the order from the POS system.
Actors	Kitchen Staff
Pre-Condition	• The order is placed successfully. • The kitchen has all the required ingredients and items for preparation.
Post-Condition	• The order is prepared, marked as completed, and served to the customer.
Result	The order is fulfilled and ready for service.
Alternative Scenario	• Kitchen prioritizes based on order time or complexity. • Customer requests a modification mid-preparation.
Exception Scenario	• Ingredient shortage causes delays. • System failure prevents the order status from being updated.
Qualities	• Efficiency: Orders are prepared within the estimated time.

Table 2.7: Order Fulfillment

Section	Content
Designation	UC-OM-03
Name	Order Tracking
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Restaurant Staff, Kitchen Portal
Responsible	Restaurant Staff, Manager
Description	Track the order status from the time it is placed until it is fulfilled. The system allows staff to monitor and update the order status at different stages.
Trigger Event	An order is placed, and its status changes (e.g., "In Progress," "Completed").
Actors	Cashier, Manager, Kitchen Staff
Pre-Condition	• The order is in the system. • The system supports real-time status updates.
Post-Condition	• The order is tracked at every stage, and the current status is visible to all relevant staff.
Result	The staff can track the progress of the order and ensure timely preparation and delivery.
Alternative Scenario	• The kitchen staff can update the status manually if there are any delays or issues. • Order priority can be modified.
Exception Scenario	• The system fails to update the order status due to technical issues. • Incorrect status is displayed, leading to delays.
Qualities	• Transparency: Clear visibility of order status.

Table 2.8: Order Tracking

Section	Content
Designation	UC-OM-04
Name	Order Cancellation
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Customer request, restaurant policy
Responsible	Cashier, Manager
Description	Allows staff or the customer to cancel an order after it has been placed but before it is fulfilled. The system handles cancellation and refund processing (if applicable).
Trigger Event	Customer requests to cancel an order, or the staff needs to cancel due to a mistake or availability issues.
Actors	Cashier, Customer, Manager
Pre-Condition	• The order has been placed but not yet fulfilled. • The cancellation policy allows the order to be canceled at this stage.
Post-Condition	• The order is canceled, and the system reflects the cancellation. • Refunds (if any) are processed.
Result	The order is successfully canceled, and the system updates accordingly.
Alternative Scenario	• Customer requests a replacement instead of cancellation. • The system prompts staff to confirm cancellation.
Exception Scenario	• The system fails to cancel the order due to technical errors. • Cancellation not allowed after a certain stage (e.g., order already being prepared).
Qualities	• Flexibility: System allows easy cancellation.

Table 2.9: Order Cancellation

Section	Content
Designation	UC-OM-05
Name	Order Modification
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Customer request, order corrections
Responsible	Cashier, Kitchen Staff
Description	Allows the Cashier to modify an order after it has been placed but before it is fulfilled. The system updates the kitchen and recalculates costs if necessary.
Trigger Event	Customer requests changes to the order, or a mistake is found in the order details.
Actors	Cashier, Kitchen Staff, Customer
Pre-Condition	• The order has been placed. • The kitchen has not started preparing the order.
Post-Condition	• The order is modified, and the kitchen is notified of the changes. • New receipt generated if applicable.
Result	The order is successfully modified, and the kitchen proceeds with the updated order.
Alternative Scenario	• The customer requests additional items or changes to the original order. • Price changes are reflected on the receipt.
Exception Scenario	• The system fails to modify the order due to technical errors. • The kitchen has already started preparing the order, and changes are not allowed.
Qualities	• Flexibility: Easy modification of orders.

Table 2.10: Order Modification

Section	Content
Designation	UC-RG-01
Name	Receipt Generation and Reprint
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	POS System, Cashier
Responsible	Cashier
Description	The cashier generates a receipt for each transaction completed in the POS system, detailing the items purchased, their prices, taxes, and total amount. Also it can reprint if change in order
Trigger Event	A customer completes their payment for an order.
Actors	Cashier, Customer
Pre-Condition	• The order is finalized, and payment is completed. • The POS system is operational.
Post-Condition	• A receipt is generated and printed. • The transaction is recorded in the system.
Result	The customer receives a receipt confirming their transaction details.
Alternative Scenario	• The cashier can choose to reprint the receipt if requested by the customer. • The system offers discounts or promotions that reflect on the receipt.
Exception Scenario	• The POS system fails to generate a receipt due to technical issues. • Items on the receipt do not match the customer's order due to input errors.
Qualities	• Accuracy: Correct details are captured and displayed on the receipt.

Table 2.11: Receipt Generation

Section	**Content**
Designation	UC-CP-01
Name	Complaint Submission
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	POS System, Complaint Portal
Responsible	Manager
Description	Managers can log customer complaints regarding service or food issues using a form within the complaint portal.
Trigger Event	A customer or staff member encounters a problem or issue during service or product delivery.
Actors	Manager
Pre-Condition	• The portal is accessible to the manager. • The system is operational for logging complaints.
Post-Condition	• The complaint is successfully logged and saved in the system for review.
Result	The complaint is stored in the system, awaiting resolution.
Alternative Scenario	• The system allows attaching images or additional information to the complaint.
Exception Scenario	• The system is down, and complaints cannot be logged. • The form fails to submit due to incomplete data or technical issues.
Qualities	• Reliability: Ensure complaints are logged without issues.

Table 2.12: Complaint Submission

Section	**Content**
Designation	UC-CP-02
Name	Complaint Review and Resolution
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Complaint Portal
Responsible	Manager, Admin
Description	Complaints logged by cashiers or staff are reviewed by the manager or admin, who assigns resolution steps or responds to the complaint.
Trigger Event	A new complaint is logged and requires review.
Actors	Manager, Admin
Pre-Condition	• The complaint is successfully submitted and logged in the system. • The manager has access to the complaint portal.
Post-Condition	• The complaint is assigned for resolution, marked as resolved, or escalated for further action.
Result	The complaint status is updated, and resolution steps are taken or recorded in the system.
Alternative Scenario	• The complaint is marked for follow-up at a later date.
Exception Scenario	• The manager cannot access the complaint system due to technical failure.
Qualities	• Accountability: Ensure proper tracking of the complaint through resolution.

Table 2.13: Complaint Review and Resolution

Section	Content
Designation	UC-CP-03
Name	Complaint Notification
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Complaint Portal
Responsible	Manager
Description	Once a complaint is logged or resolved, notifications are sent to the relevant parties (e.g., the admin)..
Trigger Event	A complaint is submitted, updated, or resolved in the system.
Actors	Manager, Customer
Pre-Condition	• A complaint is submitted and logged in the system. • The system is configured to send notifications.
Post-Condition	• Notifications are sent to the relevant parties informing them of the complaint status or next steps.
Result	All parties are informed of complaint submission, status, or resolution.
Alternative Scenario	• Notifications can be sent via multiple channels (email, SMS, or in-app). • The manager can customize the notification content or disable notifications.
Exception Scenario	• Notifications fail to be sent due to system errors. • The wrong party receives the notification.
Qualities	• Timeliness: Notifications are sent promptly.

Table 2.14: Complaint Notification

Section	Content
Designation	UC-CP-04
Name	Complaint Escalation
Author	Fatima Sarmad
Priority	Low
Criticality	Low
Source	Complaint Portal, Manager
Responsible	Manager, Admin
Description	If a complaint cannot be resolved by the first point of contact, it is escalated to higher management.
Trigger Event	A complaint is flagged as unresolved or requires escalation due to complexity or severity.
Actors	Manager, Admin
Pre-Condition	• The complaint is logged in the system. • The manager has flagged the complaint for escalation. • The resolution time for resolving at the manager's end has passed.
Post-Condition	• The complaint is transferred to senior management for further review.
Result	The complaint receives additional attention from higher-level management.
Alternative Scenario	• The complaint is escalated directly to senior management without review.
Exception Scenario	• Escalation fails due to lack of communication or system error. • The escalation is delayed, prolonging the resolution process.
Qualities	• Responsiveness: Ensure that escalations are handled quickly.

Table 2.15: Complaint Escalation

Section	Content
Designation	UC-RG-01
Name	Sales and Order Report Generation
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	POS System, Manager Portal
Responsible	Manager, Admin
Description	Generate detailed sales and order reports for the manager and admin that cover daily, weekly, and monthly sales and orders Statistics.
Trigger Event	The manager or admin requests a sales report via the system interface.
Actors	Manager, Admin
Pre-Condition	• The system contains sales and order data. • The manager/admin has access to the report generation module.
Post-Condition	• The report is generated and displayed or exported in a readable format (PDF, Excel, etc.).
Result	The manager/admin receives an up-to-date sales and order report for analysis.
Alternative Scenario	• The system allows customization of report parameters, such as specific date ranges or branch-specific sales or orders. • Automatic scheduling of reports.
Exception Scenario	• System error prevents the report from being generated. • Insufficient data for generating the requested report.
Qualities	• Accuracy: Ensure that sales and order data is accurate. • Efficiency: Reports are generated quickly and are easy to interpret.

Table 2.16: Sales and Order Report Generation

Section	Content
Designation	UC-RG-03
Name	Metrics Analysis Report
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	POS System, Manager Portal, Admin Portal
Responsible	Manager, Admin
Description	Generate detailed reports on various performance metrics such as customer satisfaction, complaint resolution rates, and stock turnover.
Trigger Event	The manager or admin requests a metrics analysis report for business insights.
Actors	Manager, Admin
Pre-Condition	• The system has sufficient data on performance metrics. • The report generation tool is available to managers or admins.
Post-Condition	• The metrics report is generated and made available for review or download.
Result	The report provides insights into business performance and operational efficiency.
Alternative Scenario	• The system can generate comparative metrics over multiple time frames or branches. • Reports are scheduled periodically (e.g., monthly).
Exception Scenario	• Some metrics are incomplete or unavailable. • The system encounters an error in compiling the data for the report.
Qualities	• Insightful: The report highlights actionable metrics.

Table 2.17: Metrics Analysis Report

Section	Content
Designation	UC-KP-01
Name	Order Completion
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Kitchen Portal, POS System
Responsible	Kitchen Staff
Description	Kitchen staff mark orders as complete once the food is prepared and ready for delivery or pickup.
Trigger Event	Kitchen staff completes the preparation of an order.
Actors	Kitchen Staff
Pre-Condition	• An order is placed and sent to the kitchen. • The kitchen staff has access to the kitchen portal.
Post-Condition	• The order is marked as complete in the system, and the status is updated in the POS system for delivery or pickup.
Result	The order is completed, and its status is updated for the cashier or staff to handle further steps.
Alternative Scenario	• The system automatically notifies the cashier or relevant staff that the order is ready for pickup or delivery.
Exception Scenario	• The kitchen staff fails to mark the order as complete due to a system error or incomplete information. • Incorrect order marked as complete.
Qualities	• Real-time Update: The order status is updated in real-time.

Table 2.18: Order Completion

Section	**Content**
Designation	UC-KP-02
Name	Active Orders View
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Kitchen Portal, POS System
Responsible	Kitchen Staff
Description	View all active orders in the kitchen, prioritize them based on order time or type, and update their statuses accordingly.
Trigger Event	Kitchen staff accesses the active orders page to check pending orders.
Actors	Kitchen Staff
Pre-Condition	• Orders are placed and sent to the kitchen. • The kitchen staff has access to the active orders view.
Post-Condition	• The kitchen staff can prioritize orders and mark them as being prepared or completed.
Result	Active orders are managed efficiently, and their statuses are updated in real time for other staff members.
Alternative Scenario	• The system can display estimated preparation times for each order. • Orders can be sorted or filtered based on priority or type of dish.
Exception Scenario	• System issues prevent new orders from being displayed. • Orders are displayed incorrectly or out of sequence.
Qualities	• Efficiency: Orders are managed smoothly to reduce wait times.

Table 2.19: Active Orders View

Section	Content
Designation	UC-KP-04
Name	Update Order Status
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Kitchen Portal, POS System
Responsible	Kitchen Staff
Description	Kitchen staff can update the status of orders as they are being prepared, in progress, or completed.
Trigger Event	Kitchen staff updates the status of an order after beginning or completing preparation.
Actors	Kitchen Staff
Pre-Condition	• The system has active orders. • Kitchen staff can access the order status update feature.
Post-Condition	• The order status is updated and reflected in the POS system for other staff members.
Result	The order status is updated in real time, ensuring the front-end staff know when an order is ready for pickup or delivery.
Alternative Scenario	• The system allows for multiple status updates (e.g., "In Progress," "Ready for Pickup," etc.).
Exception Scenario	• The system fails to update the order status due to a technical error. • Staff updates the incorrect order status by mistake.
Qualities	• Efficiency: Status updates happen in real time.

Table 2.20: Update Order Status

Section	Content
Designation	UC-WMS-01
Name	Inventory Inbound Tracking
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Warehouse Management System
Responsible	Warehouse Staff
Description	Track the incoming items into the warehouse, including details such as item type, quantity, date, and supplier information.
Trigger Event	New stock is delivered to the warehouse.
Actors	Warehouse Staff, System Administrator
Pre-Condition	• The system is operational. • Items are delivered to the warehouse. • The staff has the authority to update the inventory system.
Post-Condition	• The inbound items are recorded, and the inventory is updated accordingly in the system.
Result	Accurate record of inbound stock, helping with real-time inventory management and future forecasting.
Alternative Scenario	• The system can suggest updates for stock levels based on historical data and expected deliveries.
Exception Scenario	• The system fails to log the inbound items due to a technical error. • Incorrect quantities or item types are recorded.
Qualities	• Accuracy: Only valid and correct data is entered.

Table 2.21: Inventory Inbound Tracking

Section	Content
Designation	UC-WMS-02
Name	Inventory Outbound Tracking
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Warehouse Management System
Responsible	Warehouse Staff, System Administrator
Description	Track the outgoing items from the warehouse, including details such as item type, quantity, destination, and time of dispatch.
Trigger Event	Items are dispatched from the warehouse to other locations (e.g., branches or vendors).
Actors	Warehouse Staff, System Administrator
Pre-Condition	• The system is operational. • Items are ready for dispatch. • The staff has the authority to update the inventory system.
Post-Condition	• The outbound items are recorded, and the inventory is updated accordingly in the system.
Result	Accurate record of outbound stock, helping with real-time inventory management and tracking for replenishment.
Alternative Scenario	• The system can suggest items to be dispatched based on low stock at other branches or expected demand.
Exception Scenario	• The system fails to log the outbound items due to a technical error. • Incorrect quantities or item types are recorded.
Qualities	• Accuracy: Only valid and correct data is entered.

Table 2.22: Inventory Outbound Tracking

Section	Content
Designation	UC-WMS-03
Name	Real-time Inventory Updates
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Warehouse Management System
Responsible	Warehouse Staff, System Administrator
Description	Manage real-time updates on the stock levels of warehouse items such as ingredients or other menu inventory to ensure up-to-date tracking.
Trigger Event	Any change in the stock levels due to inbound or outbound transactions.
Actors	Warehouse Staff, System Administrator
Pre-Condition	• The system is operational. • Warehouse items are being added or removed. • Staff has access to update the system.
Post-Condition	• The system reflects updated inventory levels in real time for all items in the warehouse.
Result	Accurate real-time stock levels prevent shortages or overstock situations and improve inventory management efficiency.
Alternative Scenario	• The system can trigger automatic reordering when stock levels fall below a certain threshold. • Users can set alerts for low or high stock.
Exception Scenario	• The system fails to update stock levels in real time due to connectivity or technical issues. • Incorrect quantities are entered by mistake.
Qualities	• Efficiency: Updates occur as soon as stock is changed. • Accuracy: Precise inventory levels are maintained to avoid shortages or overstocking.

Table 2.23: Real-time Inventory Updates

Section	Content
Designation	UC-WMS-04
Name	Stock Shortages Prevention
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Warehouse Management System
Responsible	Warehouse Staff, System Administrator
Description	Monitor stock levels in real time to prevent shortages, with automated alerts for low inventory and reorder suggestions.
Trigger Event	Inventory levels fall below the minimum threshold for essential items.
Actors	Warehouse Staff, System Administrator
Pre-Condition	• Inventory levels are being tracked in real time. • Minimum stock thresholds are set in the system.
Post-Condition	• Alerts or reorder suggestions are triggered when stock falls below the minimum level.
Result	Prevent stock shortages by proactively monitoring inventory and triggering timely restocking actions.
Alternative Scenario	• System triggers automatic reorders when stock falls below critical levels.
Exception Scenario	• Alerts are not triggered in time due to system issues or incorrect threshold settings.
Qualities	• Proactivity: Stock shortages are avoided through early alerts.

Table 2.24: Stock Shortages Prevention

Section	Content
Designation	UC-WMS-05
Name	Excess Stock Management
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Warehouse Management System
Responsible	Warehouse Staff, System Administrator
Description	Monitor stock levels to avoid excess inventory and recommend actions such as discounts or redistribution to other branches.
Trigger Event	Inventory levels exceed the maximum threshold set for certain items.
Actors	Warehouse Staff, System Administrator
Pre-Condition	• Real-time tracking of stock levels is enabled. • Maximum stock thresholds are set in the system.
Post-Condition	• Recommendations are generated for dealing with excess inventory (e.g., redistribute, apply discounts).
Result	Prevent overstocking by generating actionable insights for dealing with excess inventory.
Alternative Scenario	• System automatically suggests redistributing items to branches with lower stock levels. • Discounts are suggested for overstocked items.
Exception Scenario	• System fails to track excess stock accurately due to incorrect input or technical errors. • Overstocked items are not managed in time.
Qualities	• Responsiveness: Immediate action is suggested for excess stock.

Table 2.25: Excess Stock Management

Section	Content
Designation	UC-WMS-06
Name	Demand Forecasting
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Warehouse Management System
Responsible	System Administrator, Warehouse Manager
Description	Use automated alerts and predictive models to forecast future demand for inventory items based on historical sales data, seasonality, and other data sources.
Trigger Event	Sales data and other relevant information are updated, prompting the system to generate forecasts.
Actors	Warehouse Manager, System Administrator
Pre-Condition	• Historical sales data and other relevant metrics are available. • Predictive models are in place. • System is operational.
Post-Condition	• System generates accurate demand forecasts and sends automated alerts to relevant actors for stock planning.
Result	Inventory is managed efficiently to meet predicted demand, reducing both shortages and excess stock.
Alternative Scenario	• The system can suggest seasonal stock adjustments or special inventory requirements based on upcoming events or promotions.
Exception Scenario	• Forecasting model fails to generate accurate predictions due to data discrepancies or model errors. • Alerts are not triggered in time.
Qualities	• Efficiency: Alerts are sent in time.

Table 2.26: Demand Forecasting

Section	**Content**
Designation	UC-WMS-07
Name	Automated Alerts
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Warehouse Management System
Responsible	System Administrator, Warehouse Manager
Description	System sends automated alerts to notify warehouse managers about forecasted demand spikes or shortages.
Trigger Event	Forecasted demand exceeds or falls below predefined thresholds, triggering an alert.
Actors	Warehouse Manager, System Administrator
Pre-Condition	• Demand forecasting is enabled. • Notification thresholds are defined in the system.
Post-Condition	• Alerts are generated and sent to responsible actors for stock adjustments.
Result	Timely alerts allow warehouse staff to prepare for potential stock shortages or overstock situations.
Alternative Scenario	• Alerts can be customized based on urgency or category (e.g., high-priority items).
Exception Scenario	• Alerts are not sent due to system failure or incorrect configuration of alert thresholds.
Qualities	• Timeliness: Alerts are sent well in advance of potential stock issues.

Table 2.27: Automated Alerts

Section	Content
Designation	UC-WMS-08
Name	Predictive Models
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Warehouse Management System
Responsible	System Administrator
Description	Utilize machine learning models to predict future demand based on historical sales trends, seasonality, and external data sources such as market conditions.
Trigger Event	Sales and external data sources are updated, and the model generates new demand predictions.
Actors	Warehouse Manager, System Administrator
Pre-Condition	• Historical data is available. • The machine learning model is trained and operational.
Post-Condition	• Accurate demand predictions are generated, and relevant staff are notified.
Result	Warehouse managers receive actionable insights to adjust stock levels based on future demand predictions.
Alternative Scenario	• The system can suggest model retraining based on new data patterns or external factors (e.g., market disruptions).
Exception Scenario	• Predictive model fails to generate accurate predictions due to incorrect data inputs or insufficient training data.
Qualities	• Accuracy: Predictions are based on comprehensive data and fine-tuned models.

Table 2.28: Predictive Models

Section	Content
Designation	UC-MP-01
Name	Branch Management - CRUD Operations
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Manager Portal
Responsible	System Administrator, Manager
Description	Manage the operations of multiple restaurant branches, including creating, reading, updating, and deleting branch records. Managers can view and control branch details, including sales, stock levels, and customer feedback, all from a centralized portal.
Trigger Event	Manager logs into the portal and selects the branch management option.
Actors	Manager, System Administrator
Pre-Condition	• The manager is logged into the system. • Branches are registered in the system. • The manager has the required permissions.
Post-Condition	• Branch records are created, updated, deleted, or retrieved successfully. • Changes reflect immediately across all modules, such as sales and stock levels.
Result	Managers can efficiently manage multiple restaurant branches, ensuring visibility and control over critical branch operations.
Alternative Scenario	• The system allows bulk updates for managing multiple branches simultaneously. • Managers can export branch data for external analysis.
Exception Scenario	• System errors occur while creating or updating branches. • Insufficient permissions prevent the manager from performing the required operations.

Table 2.29: Branch Management - CRUD Operations

Section	Content
Designation	UC-MP-02
Name	Cashier and Staff Management
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Manager Portal
Responsible	System Administrator, Manager
Description	Manage restaurant staff including cashiers, kitchen staff, and warehouse personnel. The system allows adding, updating, viewing, and removing staff records, assigning shifts, and defining roles and responsibilities.
Trigger Event	Manager accesses the staff management feature from the portal.
Actors	Manager, System Administrator
Pre-Condition	• The manager is logged into the system. • Staff data exists in the system. • The manager has the required permissions.
Post-Condition	• Staff records are created, updated, deleted, or retrieved successfully. • Staff shifts and roles are managed effectively.
Result	The manager efficiently manages staff details, including roles, shifts, and performance, ensuring smooth operations across the restaurant.
Alternative Scenario	• The system allows bulk management for assigning roles and shifts to multiple staff members at once. • Shift management supports automated scheduling based on predefined staff availability.
Exception Scenario	• System errors occur while managing staff records (e.g., network failure). • Insufficient permissions prevent the manager from accessing staff details.
Qualities	• Efficiency: Staff management operations are performed swiftly.

Table 2.30: Cashier and Staff Management

Section	Content
Designation	UC-MP-03
Name	CRUD Operations for Cashier and Staff Management
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Manager Portal
Responsible	System Administrator, Manager
Description	Create, read, update, and delete staff records, including cashier and staff roles. Manage shifts and staff responsibilities from a centralized portal.
Trigger Event	Manager selects a CRUD operation for managing staff (e.g., create new staff, update shift, delete record).
Actors	Manager, System Administrator
Pre-Condition	• The manager is logged in. • The manager has the required permissions. • Staff data is available in the system.
Post-Condition	• CRUD operations on staff records are executed successfully. • Shifts and roles are updated in real time across relevant modules.
Result	Staff management operations, including hiring, firing, shift scheduling, and role assignments, are streamlined for restaurant management.
Alternative Scenario	• Bulk staff record updates for changes in shift or role assignments. • Automated shift planning based on staff availability and operational needs.
Exception Scenario	• Errors in staff data entry (e.g., invalid details) or system errors prevent the manager from completing the operation.

Table 2.31: CRUD Operations for Cashier and Staff Management

Section	Content
Designation	UC-MP-04
Name	Shift Management
Author	Fatima Sarmad
Priority	High
Criticality	Medium
Source	Manager Portal
Responsible	Manager
Description	Allows managers to assign, modify, and manage staff shifts across different branches. It ensures each shift is properly covered, prevents scheduling conflicts, and supports shift swapping when necessary.
Trigger Event	Manager logs in and selects the staff shift management feature.
Actors	Manager, Staff
Pre-Condition	• Staff data is available. • Shift schedules are defined in the system. • The manager has the necessary permissions to manage shifts.
Post-Condition	• Staff shifts are updated, conflicts are resolved, and any necessary notifications are sent to staff about shift changes.
Result	Shift management is handled efficiently, ensuring smooth restaurant operations and coverage for all shifts.
Alternative Scenario	• Auto-scheduling allows the system to automatically assign shifts based on staff availability and historical data. • Shift changes or swaps are requested and processed by staff themselves.
Exception Scenario	• Scheduling conflicts (e.g., two staff members scheduled for the same shift) prevent the manager from finalizing the shift plan.
Qualities	• Efficiency: The system allows quick resolution.

Table 2.32: Shift Management

Section	Content
Designation	UC-MP-05
Name	Performance Tracking
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Manager Portal
Responsible	Manager
Description	Allows managers to track the performance of staff based on KPIs such as customer feedback, and sales performance for cashiers. This use case helps in making informed decisions for promotions or shifts adjustments.
Trigger Event	Manager logs into the portal and views staff performance metrics.
Actors	Manager, Staff
Pre-Condition	• Staff data is available. • Performance metrics have been defined in the system. • Historical performance data is stored.
Post-Condition	• The manager views the performance report for staff. • Performance data is used for decision-making (e.g., shifts, bonuses, or promotions).
Result	Performance reports allow managers to improve operations by rewarding or managing staff more effectively.
Alternative Scenario	• Integration with customer feedback allows real-time updates to staff performance based on customer satisfaction. • Alerts for underperforming staff.
Exception Scenario	• Missing performance data results in incomplete reports, preventing managers from making informed decisions.

Table 2.33: Performance Tracking

Section	Content
Designation	UC-MP-06
Name	Role Assignment
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Manager Portal
Responsible	Manager
Description	Allows managers to assign specific roles to staff members (e.g., cashier, kitchen staff) and change roles when necessary. Each role has specific responsibilities and permissions within the system.
Trigger Event	Manager logs in and selects a staff member to assign a role.
Actors	Manager, Staff
Pre-Condition	• Staff members are registered in the system. • Roles are predefined in the system. • The manager has the necessary permissions.
Post-Condition	• Staff roles are updated or assigned successfully in the system. • Staff permissions change based on their new role.
Result	Roles are assigned, ensuring that staff members are given the correct responsibilities and access within the system.
Alternative Scenario	• The system allows managers to create custom roles for staff based on specific needs of the branch. • Automatic role changes based on staff performance or tenure.
Exception Scenario	• Errors in assigning roles prevent staff from performing their duties. • Insufficient permissions prevent the manager from completing the role assignment.

Table 2.34: Role Assignment

Section	Content
Designation	UC-MP-07
Name	Recommendation and Feedback Review
Author	Fatima Sarmad
Priority	High
Criticality	Medium
Source	Manager Portal
Responsible	Manager, System Administrator
Description	Allows the manager to review customer feedback and receive system-generated recommendations. This feedback can come from various sources.
Trigger Event	Manager logs into the system and selects the feedback and recommendation section.
Actors	Manager
Pre-Condition	• Feedback data is available in the system. • Recommendation algorithms are set up and functional. • The manager is logged in and has the necessary permissions.
Post-Condition	• The manager reviews feedback and receives actionable recommendations. • Feedback analysis is completed, and the system generates suggestions.
Result	The restaurant's operations, services, and products are adjusted based on the feedback and recommendations.
Alternative Scenario	• The system groups feedback based on categories like food quality, service, or ambiance, allowing managers to focus on specific areas for improvement.
Exception Scenario	• Feedback or recommendation data is incomplete or missing, preventing the system from generating actionable recommendations.

Table 2.35: Recommendation and Feedback Review

Section	Content
Designation	UC-MP-08
Name	Feedback Analysis
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Manager Portal
Responsible	Manager
Description	The system analyzes customer feedback to generate detailed reports. This helps the manager focus on addressing the most critical customer concerns.
Trigger Event	Feedback is collected from different sources and analyzed by the system.
Actors	Manager
Pre-Condition	• Feedback is available from multiple channels. • The system's feedback analysis feature is operational. • The manager is logged in and authorized to view reports.
Post-Condition	• The feedback analysis is completed, and a report is generated. • The manager reviews the analysis and identifies areas for improvement.
Result	Insights from the feedback analysis inform the manager's decisions, leading to better service.
Alternative Scenario	• The system provides visual insights (charts, graphs) for feedback trends, helping managers quickly grasp key findings. • Integration with external review platforms allows for more comprehensive analysis.
Exception Scenario	• Missing or insufficient data results in inaccurate or incomplete feedback analysis reports. • Feedback categories are too general.

Table 2.36: Feedback Analysis

Section	**Content**
Designation	UC-MP-09
Name	Recommendation System
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Manager Portal
Responsible	Manager, System Administrator
Description	A recommendation system analyzes feedback, customer preferences, and sales data to suggest operational improvements. Recommendations may also include offers, changes in pricing, or staff training.
Trigger Event	The system automatically generates recommendations based on the analysis of feedback, sales, and customer trends.
Actors	Manager, System Administrator
Pre-Condition	• Feedback and sales data are available for analysis. • Recommendation algorithms are functional. • The manager has access to the recommendation system.
Post-Condition	• Recommendations are generated and displayed to the manager. • The manager reviews and implements actionable suggestions.
Result	The manager implements suggested changes to improve restaurant performance.
Alternative Scenario	• Recommendations are categorized by urgency or impact, helping managers prioritize actions. • Recommendations update dynamically as new data is collected.
Exception Scenario	• The recommendation system fails to generate accurate suggestions due to missing or incomplete data.

Table 2.37: Recommendation System

Section	Content
Designation	UC-AP-01
Name	Predictive Sales Analysis
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Administration Portal
Responsible	Administrator
Description	The system analyzes historical sales data to generate predictive models that forecast future sales trends. This helps administrators plan for upcoming demand, promotions, and stock management.
Trigger Event	Administrator requests a sales forecast report from the system.
Actors	Administrator
Pre-Condition	• Historical sales data is available in the system. • The predictive analysis feature is functional and trained on relevant datasets. • The administrator is logged in and has the necessary permissions.
Post-Condition	• A sales forecast report is generated. • The administrator reviews and uses the data to make informed decisions about inventory, staffing, and promotions.
Result	The sales forecast helps improve operational planning, ensuring that the restaurant is prepared for future demand.
Alternative Scenario	• The system generates multiple sales forecasts based on different assumptions, such as peak season or promotional periods.
Exception Scenario	• Inaccurate or incomplete sales data results in flawed predictions, requiring manual review or adjustments to the predictive model.

Table 2.38: Predictive Sales Analysis

Section	Content
Designation	UC-AP-02
Name	Predictive Stock Needs
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Administration Portal
Responsible	Administrator, Warehouse Manager
Description	The system uses sales data, seasonality trends, and customer preferences to predict future stock requirements. This helps ensure the restaurant maintains optimal stock levels without overordering or understocking.
Trigger Event	Administrator or Warehouse Manager requests a stock forecast based on predicted sales.
Actors	Administrator, Warehouse Manager
Pre-Condition	• Historical stock data and sales trends are available. • The predictive analysis tool is operational. • The actors have appropriate permissions.
Post-Condition	• The system generates a stock needs report, forecasting how much inventory will be required to meet predicted sales demand.
Result	Stock levels are optimized based on sales forecasts, reducing waste and ensuring product availability.
Alternative Scenario	• The system provides detailed stock reports, broken down by product category or location, allowing for more granular control.
Exception Scenario	• The stock prediction model fails due to incorrect or missing input data, leading to either overstocking or understocking issues.
Qualities	• Efficiency: Helps prevent stock shortages and reduces excess inventory.

Table 2.39: Predictive Stock Needs

Section	Content
Designation	UC-AP-03
Name	Customer Trend Prediction
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Administration Portal
Responsible	Administrator
Description	The system analyzes customer feedback, preferences, and market trends to predict future customer demands. These predictions help restaurant managers tailor menu items, promotions, and services to align with evolving customer preferences.
Trigger Event	Administrator requests a customer trend analysis report.
Actors	Administrator, Warehouse Manager
Pre-Condition	• Historical customer data and behavior trends are available. • The system is configured to perform customer trend analysis. • The administrator is logged in with the necessary permissions.
Post-Condition	• A customer trend report is generated and reviewed by the administratorto plan future strategies.
Result	The restaurant adapts its offerings to meet predicted customer trends, increasing customer satisfaction and sales.
Alternative Scenario	• The system offers trend predictions for different segments, such as customer demographics or seasonality, allowing for more targeted marketing efforts.
Exception Scenario	• The system provides conflicting predictions, or customer data is too sparse to generate meaningful trends, leading to less reliable insights.

Table 2.40: Customer Trend Prediction

Section	Content
Designation	UC-CA-01
Name	Competitor Price Comparison
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Administration Portal
Responsible	Administrator
Description	The system compares competitor prices across multiple localities, giving restaurant administrators an overview of how their prices stand against competitors. This helps in pricing strategies and promotions.
Trigger Event	Administrator requests a competitor price comparison report.
Actors	Administrator
Pre-Condition	• Competitor data is available through web scraping. • The system is getting data from FoodPanda. • Administrator is logged in with proper access.
Post-Condition	• A detailed report on competitor prices is generated. • The team uses the report to adjust pricing strategies for specific locations.
Result	Restaurant prices are adjusted to remain competitive, enhancing the ability to attract customers.
Alternative Scenario	• The system offers segmented price comparison reports based on region, type of restaurant, or item categories, allowing for more targeted pricing adjustments.
Exception Scenario	• Failure to retrieve competitor data due to issues with web scraping or data source availability results in incomplete or inaccurate comparisons.
Qualities	• Accuracy: Prices are compared based on real-time data from relevant competitors.

Table 2.41: Competitor Price Comparison

Section	Content
Designation	UC-CA-02
Name	Competitor Deal Comparison
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Administration Portal
Responsible	Administrator
Description	The system analyzes competitor deals and promotional offers, providing the restaurant administrators with insights into ongoing promotions in the market. This helps with crafting competitive deals.
Trigger Event	Administrator requests a competitor deal analysis.
Actors	Administrator
Pre-Condition	• Competitor deal information is available. • The system is set up to collect and analyze deal data from FoodPanda. • Administrator is logged in with necessary permissions.
Post-Condition	• A report detailing competitor deals and promotions is generated. • The team uses this data to create competitive promotions.
Result	Competitive deals and promotions are designed, helping the restaurant attract more customers.
Alternative Scenario	• The system provides promotional data broken down by deal type (e.g., discounts, buy-one-get-one-free offers), allowing for more strategic promotion planning.
Exception Scenario	• Incomplete or outdated data on competitor deals results in flawed insights, requiring manual adjustment or supplementary data sources.
Qualities	• Timeliness: Deals are updated regularly.

Table 2.42: Competitor Deal Comparison

Section	Content
Designation	UC-CA-03
Name	Customer Preference Analysis
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Administration Portal
Responsible	Administrator
Description	The system analyzes competitor customer reviews and preferences to understand what drives customer positive feedback. This helps restaurant administrators fine-tune their offerings and services.
Trigger Event	Administrator requests a customer preference analysis based on competitor data.
Actors	Administrator
Pre-Condition	• Customer reviews and preference data from competitors are available. • The system is set up to scrape or collect this data from sources like FoodPanda.pk.
Post-Condition	• A report on customer preferences and pain points is generated, highlighting competitor strengths and weaknesses. • The team uses the report to refine the restaurant's offerings.
Result	The restaurant adapts its services and menu offerings based on insights from competitor customer preferences, improving customer satisfaction.
Alternative Scenario	• The system offers segmented reports based on customer demographics or geographic areas, providing more tailored insights.
Exception Scenario	• Insufficient or unreliable competitor data results in less actionable insights, requiring manual review or adjustments to the analysis model.

Table 2.43: Customer Preference Analysis

Section	Content
Designation	UC-RS-01
Name	Menu Recommendation
Author	Fatima Sarmad
Priority	High
Criticality	High
Source	Administration Portal
Responsible	Administrator
Description	The system analyzes customer behavior and market trends to offer recommendations on adding or removing items from the menu.
Trigger Event	Administrator requests menu optimization based on customer preferences or market trends.
Actors	Administrator
Pre-Condition	• Customer behavior data is available. • Market trend data is collected regularly. • The system is integrated with feedback analysis tools.
Post-Condition	• Menu recommendations are generated, helping optimize the offerings. • The team uses the insights to update the menu.
Result	A more tailored menu that aligns with customer preferences and market trends, improving customer satisfaction and sales.
Alternative Scenario	• The system provides segmented menu recommendations based on customer demographics, locations, or time of year (e.g., seasonal dishes).
Exception Scenario	• Lack of sufficient customer behavior or market trend data results in less accurate recommendations, requiring manual review or adjustment by the team.

Table 2.44: Menu Recommendation

Section	Content
Designation	UC-RS-02
Name	Pricing Adjustment Recommendation
Author	Fatima Sarmad
Priority	Medium
Criticality	High
Source	Administration Portal
Responsible	Administrator
Description	The system recommends price adjustments based on customer purchase behavior, competitor pricing, and market trends. This helps maintain competitiveness and maximize revenue.
Trigger Event	Administrator requests pricing recommendations based on current market conditions or competitor analysis.
Actors	Administrator
Pre-Condition	• Customer purchasing behavior and competitor pricing data are available. • The system is set up to gather market trends and competitor price data.
Post-Condition	• Pricing recommendations are generated, and the team updates prices accordingly.
Result	Prices are optimized for market competitiveness and customer preferences, helping increase sales and profit margins.
Alternative Scenario	• The system offers dynamic pricing suggestions, adjusting based on time of day, seasonality, or promotions.
Exception Scenario	• Incomplete competitor data or outdated market trends lead to inaccurate pricing suggestions, requiring manual adjustments.

Table 2.45: Pricing Adjustment Recommendation

Section	**Content**
Designation	UC-RS-03
Name	Promotional Deal Recommendation
Author	Fatima Sarmad
Priority	Medium
Criticality	Medium
Source	Administration Portal
Responsible	Administrator
Description	The system suggests promotional deals and offers based on customer behavior, purchase patterns, and competitor promotions, helping attract more customers.
Trigger Event	Administrator requests promotional recommendations based on current customer trends or competitor promotions.
Actors	Administrator
Pre-Condition	• Customer purchase data and competitor promotion data are available. • The system is connected to promotional trend data and feedback analysis.
Post-Condition	• Promotional deal suggestions are generated, and the team implements the offers in the system.
Result	The restaurant attracts more customers with well-targeted promotional offers, improving sales and customer retention.
Alternative Scenario	• The system offers segmented promotional deal suggestions based on customer demographics or purchase history.
Exception Scenario	• Insufficient customer or competitor data results in less effective promotional recommendations, requiring manual adjustments.
Qualities	• Effectiveness: Ensuring promotional deals attract the target audience.

Table 2.46: Promotional Deal Recommendation

2.1 Functional Requirements

Functional requirements define what a product must do and what its features and functions are. Following are the functional requirements for our project:

2.1.1 Point of Sale (POS) System

1. The system shall allow staff to manage customer orders, add items to carts, and process payments.
2. The system shall track the entire order life cycle from placing an order to kitchen delivery and order completion.
3. The system shall generate receipts for completed transactions after payment is processed.
4. The system shall allow staff to log and manage complaints regarding service or product issues through the complaint portal.
5. The system shall provide detailed reports on sales, orders, and customer metrics for managers and administrators.

2.1.2 Customer Management

1. The system shall track customer order history and preferences.
2. The system shall provide customer profiles that store past interactions and dining preferences to personalize their experience.
3. The system shall allow staff to view and update customer information.

2.1.3 Order Management

1. The system shall manage the workflow of orders, including tracking orders sent to the kitchen and their fulfillment status.
2. The system shall allow staff to update the status of an order (e.g., placed, in progress, completed).
3. The system shall notify kitchen staff when a new order is placed.

2.1.4 Receipt Generation

1. The system shall generate a receipt for every transaction processed by the cashier.
2. The system shall allow the cashier to print the reciept.
3. The system shall track receipts in the database for future reference and reporting.

2.1.5 Complaint Portal

1. The system shall allow staff and customers to log complaints about service or product issues.
2. The system shall allow managers to view and resolve complaints.
3. The system shall maintain a record of complaints for quality improvement analysis.

2.1.6 Report Generation

1. The system shall generate detailed reports on sales, order processing times, and stock management.
2. The system shall allow managers to view daily, weekly, monthly, and custom period reports.
3. The system shall provide visual reports (graphs, charts) for better insight.

2.1.7 Kitchen Portal

1. The system shall allow kitchen staff to view all active orders and prioritize them.
2. The system shall allow kitchen staff to update the status of orders (e.g., in preparation, ready for delivery).
3. The system shall allow kitchen staff to mark orders as complete when fulfilled.

2.1.8 Warehouse Management

1. The system shall track the inbound and outbound flow of stock items.
2. The system shall allow real-time updates on inventory levels to avoid stock shortages or excesses.

3. The system shall generate automated alerts and predictions for future stock needs based on historical sales data.

2.1.9 Manager Portal

1. The system shall allow managers to manage restaurant branches, including sales data and stock visibility.
2. The system shall provide tools for managing staff, their shifts, and their responsibilities.
3. The system shall allow managers to add, update, or remove products from the POS system.
4. The system shall enable managers to review feedback and provide operational recommendations based on customer input.

2.1.10 Administration Portal

1. The system shall generate predictive models for sales, future stock needs, and customer trends.
2. The system shall allow administrators to compare prices, deals, and customer preferences of competitors in the locality.
3. The system shall provide a recommender system for targeted changes to menu items, pricing adjustments, and promotional deals.

2.1.11 Web Scraper for Competitive Analysis

1. The system shall scrape competitor data (e.g., menus, prices, reviews) from platforms like Foodpanda.pk.
2. The system shall collect data from various cities and restaurants for competitive and predictive analysis.
3. The system shall provide an API for integrating scraped data into predictive models.

2.1.12 Predictive Analytics

1. The system shall generate sales forecasts based on historical data and market trends.

2. The system shall analyze customer preferences based on regions, events, and seasonality.
3. The system shall recommend operational efficiency improvements based on predictive models.

2.1.13 Feedback Analysis

1. The system shall collect and analyze customer feedback.
2. The system shall perform analysis to identify trends in customer satisfaction.
3. The system shall provide tailored recommendations to improve the customer experience based on feedback.

2.2 Non-Functional Requirements

Nonfunctional requirements describe the general properties of a system. They are also known as quality attributes. Following are the non-functional requirements for our project:

2.2.1 Performance

1. The system shall process customer orders and update inventory in real-time, with a maximum delay of 2 seconds for database updates.
2. The system shall generate reports and predictive models within 10 seconds of request submission.

2.2.2 Scalability

1. The system shall scale to accommodate up to 100 restaurant branches with independent POS systems.
2. The system shall handle the addition of new users and restaurant branches without significant performance degradation.

2.2.3 Reliability

1. The system shall maintain 99.9% uptime to ensure continuous availability for order management and reporting.

2. The system shall ensure that all transaction data (e.g., orders, receipts) are securely stored and not lost during system crashes.

2.2.4 Usability

1. The system shall provide an intuitive user interface for staff, managers, and administrators, with minimal training required for operation.

2. The system shall provide role-based access controls, ensuring that only authorized users can access certain functionalities (e.g., manager-level reporting, administration-level predictive analysis).

2.2.5 Security

1. The system shall ensure secure handling of customer data, complying with applicable data protection regulations (e.g., GDPR, local regulations).

2. The system shall provide user authentication and role-based access control to ensure sensitive data is only accessible to authorized users.

3. The system shall securely transmit data across branches and portals using encryption.

2.2.6 Maintainability

1. The system shall allow easy updates to add new features, fix bugs, or enhance system performance.

2. The system shall support continuous monitoring for errors, with automated alerts for system issues.

3. The system shall provide comprehensive documentation for developers and IT staff.

2.2.7 Data Integrity

1. The system shall ensure that customer orders, inventory data, and financial transactions are accurately recorded and synchronized across branches.

2. The system shall maintain audit logs for all transactions, manual overrides, and data modifications for future reference.

2.2.8 Compliance

1. The system shall adhere to local regulations regarding financial transactions, tax calculations, and data privacy.
2. The system shall support the generation of compliance reports for audit purposes.

2.2.9 Availability

1. The system shall be available 24/7, with a scheduled maintenance window that does not exceed 2 hours per month.
2. The system shall have a disaster recovery plan in place to ensure rapid recovery in case of system failure.

2.2.10 Responsiveness

1. The system shall provide feedback to users within 2 seconds of user actions, such as submitting an order or generating a report.
2. The system shall notify users in real time of any issues with stock levels or orders through alerts.

2.2.11 Usability

Usability requirements deal with ease of learning, ease of use, error avoidance and recovery, the efficiency of interactions, and accessibility. The usability requirements specified here will help the user interface designer create the optimum user experience. Example:

USE-1: The COS shall allow a user to retrieve the previous meal ordered with a single interaction.

2.2.12 Performance

State specific performance requirements for various system operations. If different functional requirements or features have different performance requirements, it's appropriate to specify those performance goals right with the corresponding functional requirements, rather than collecting them in this section. Example:

PER-1: 95 percent of webpages generated by the COS shall download completely within 4 seconds from the time the user requests the page over a 20 Mbps or faster Internet connection.

Chapter 3

System Overview

3.1 Architectural Design

3.1.1 Activity Diagrams

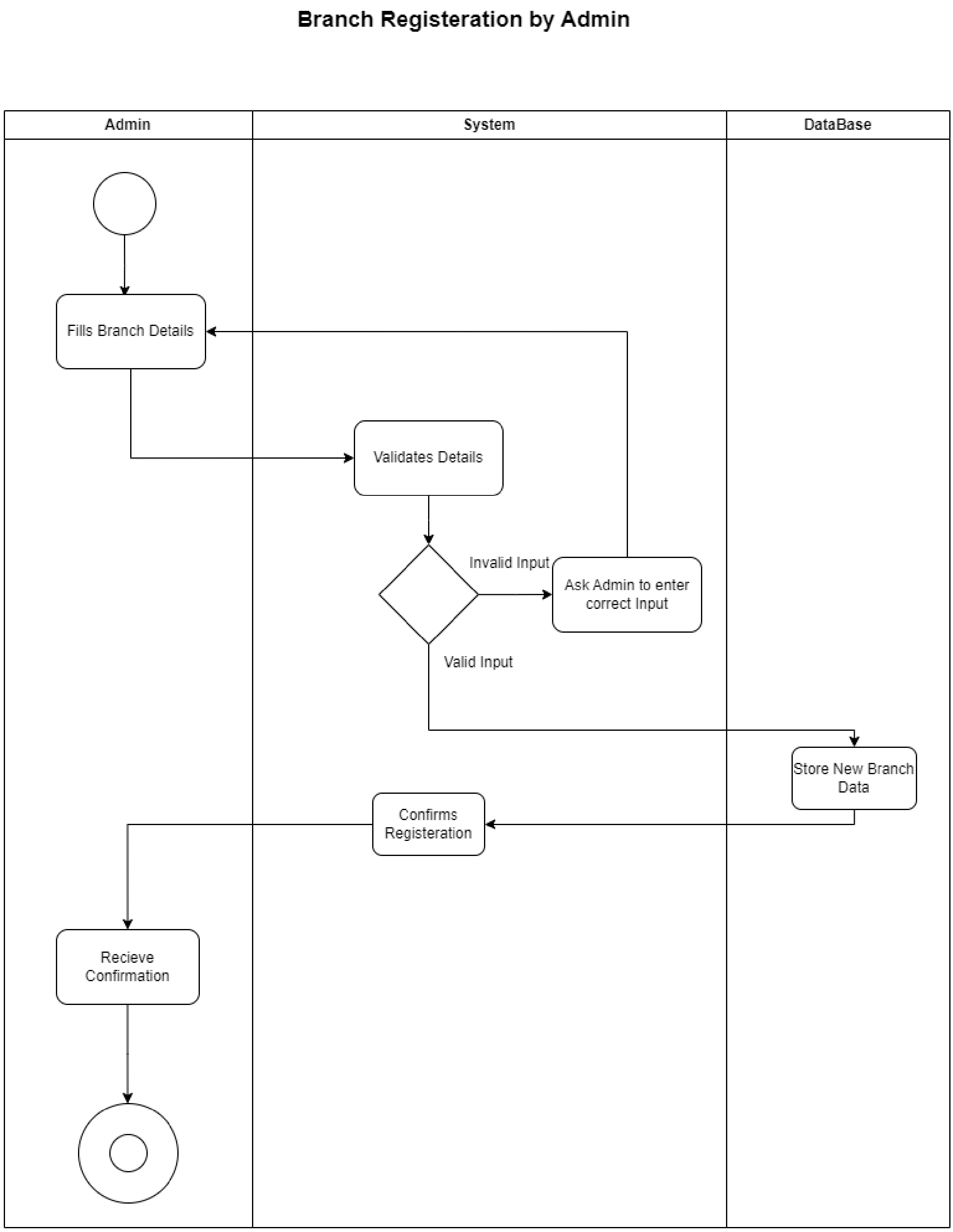

Figure 3.1: Branch Registration by Admin

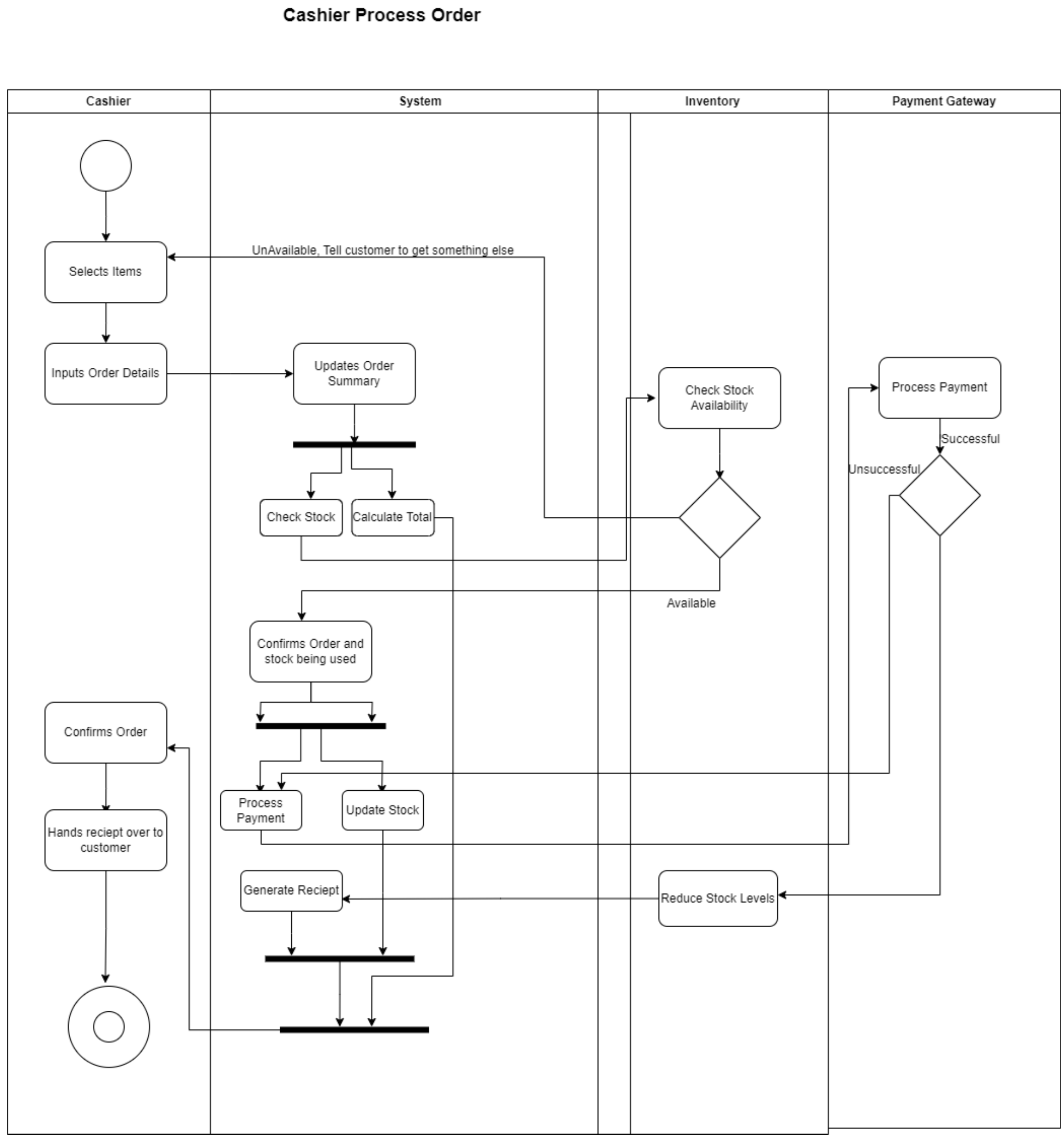

Figure 3.2: Cashier Process Order

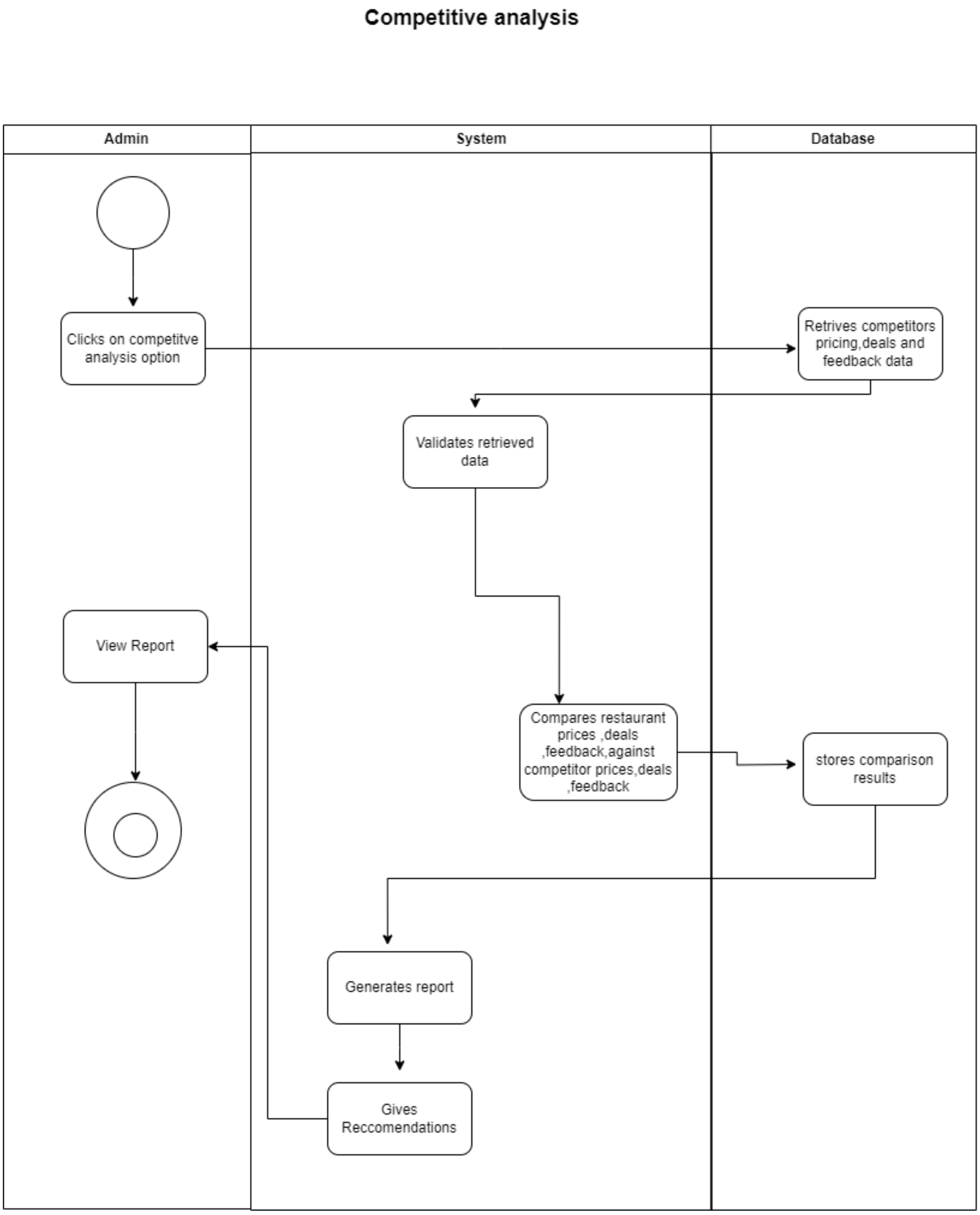

Figure 3.3: Competitive Analysis

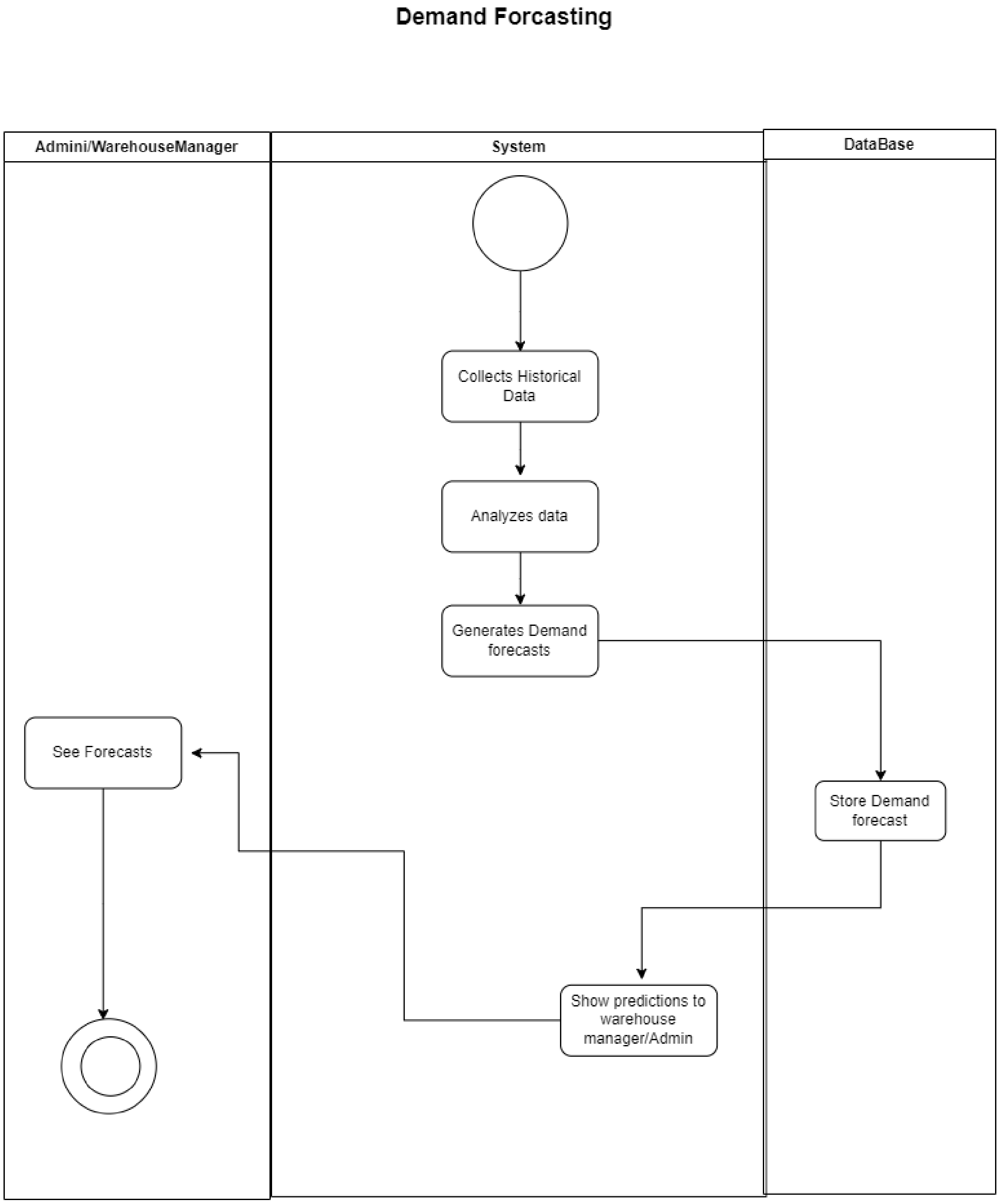

Figure 3.4: Demand Forecasting

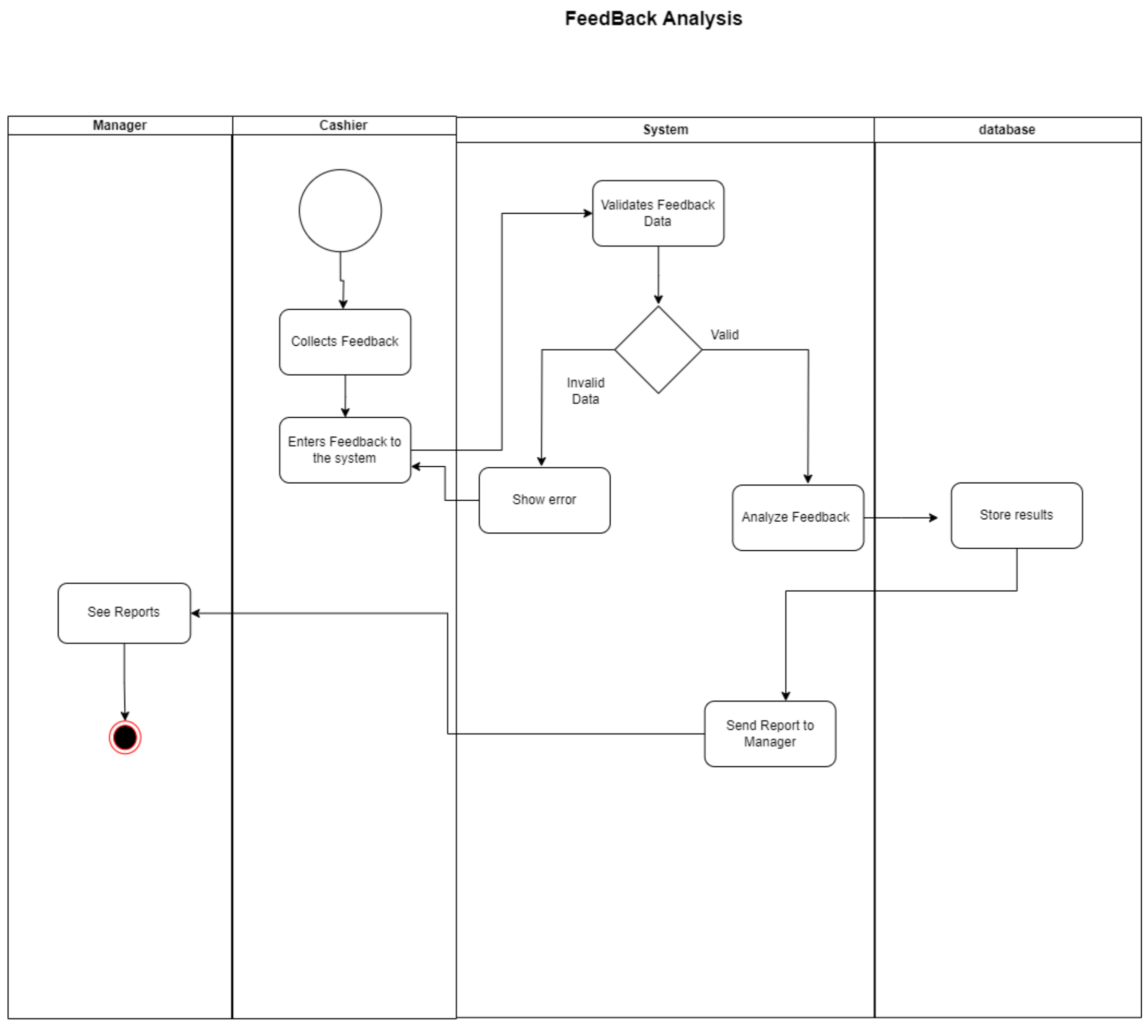

Figure 3.5: Feedback Analysis

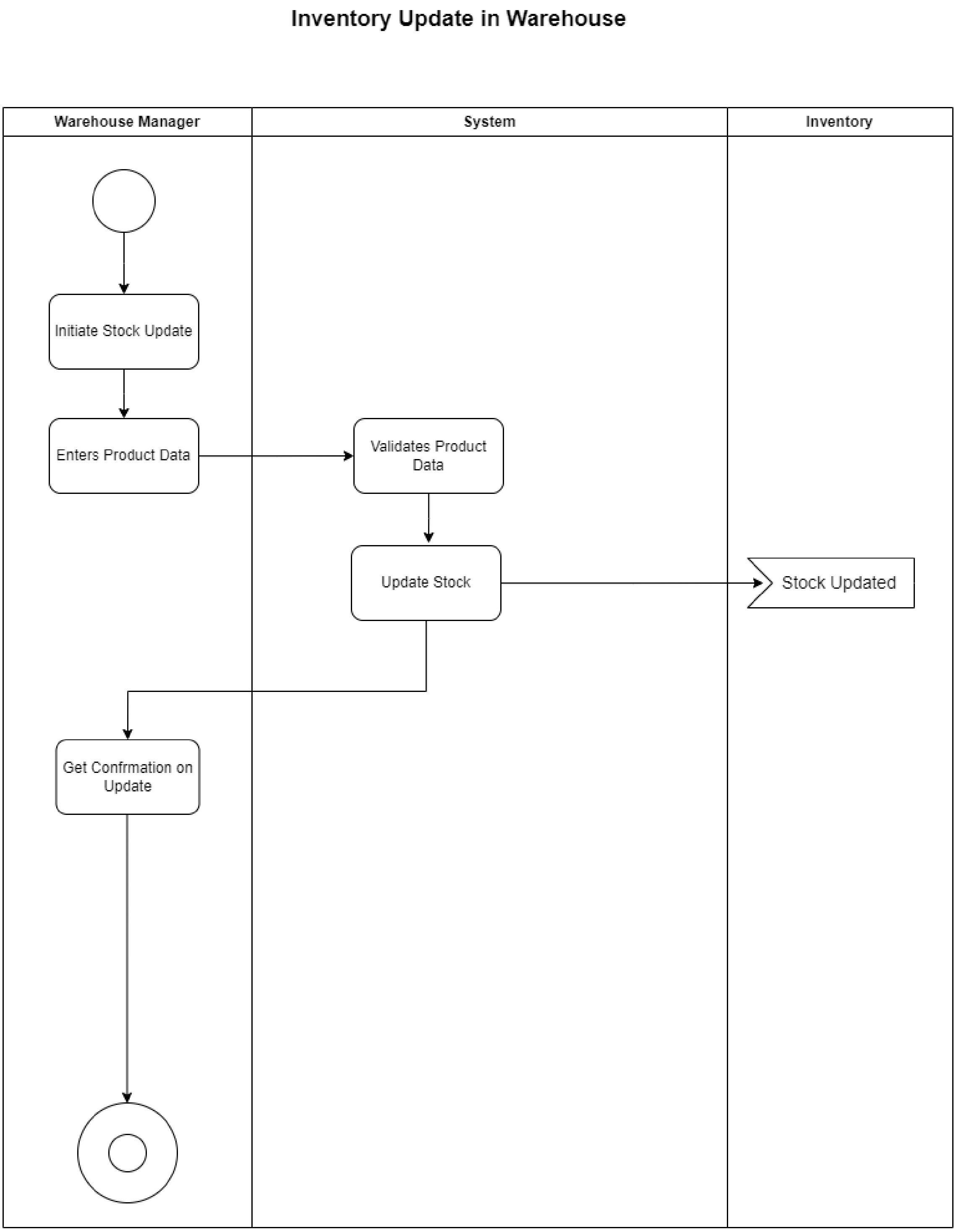

Figure 3.6: Inventory Update in Warehouse

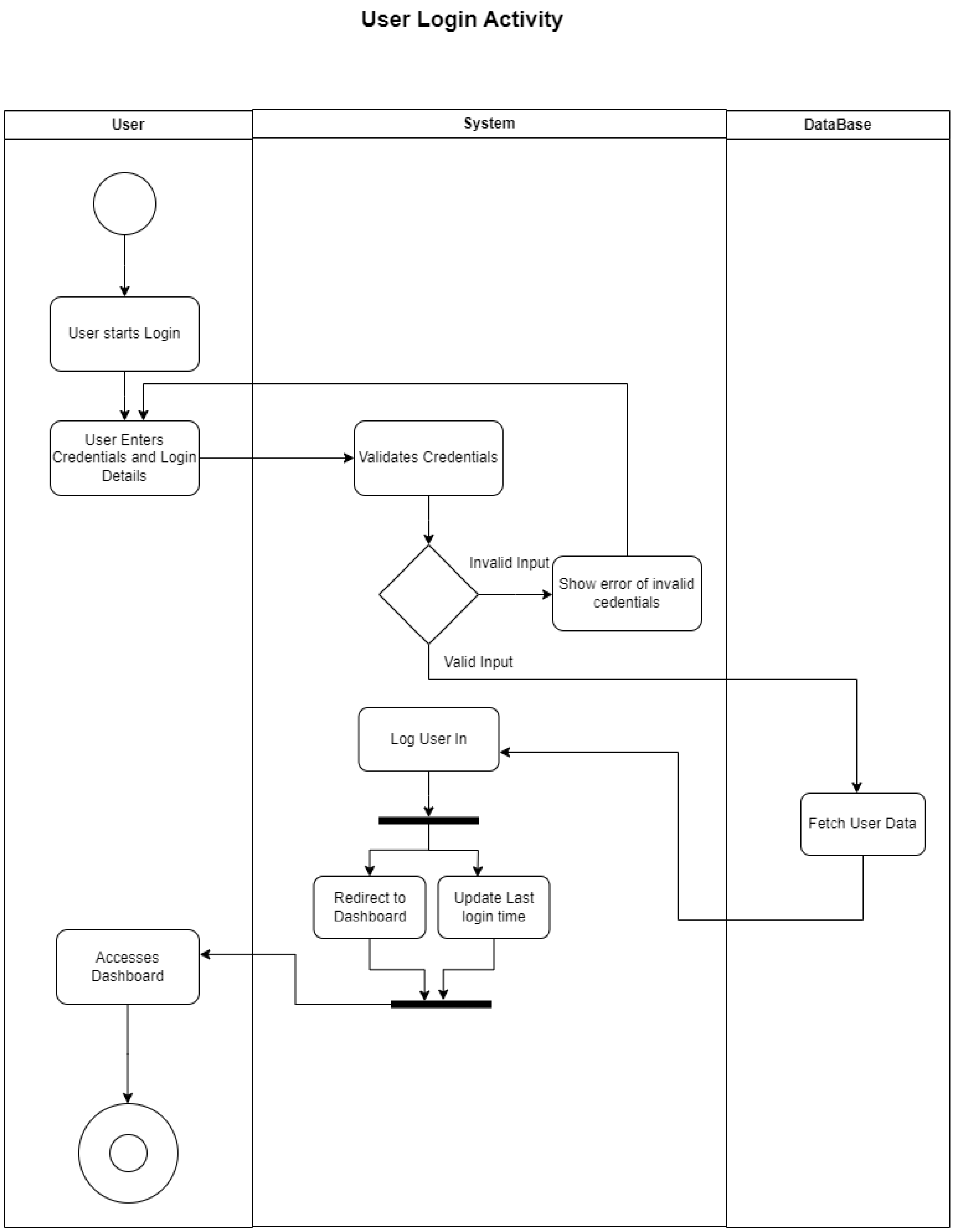

Figure 3.7: Log in Activity

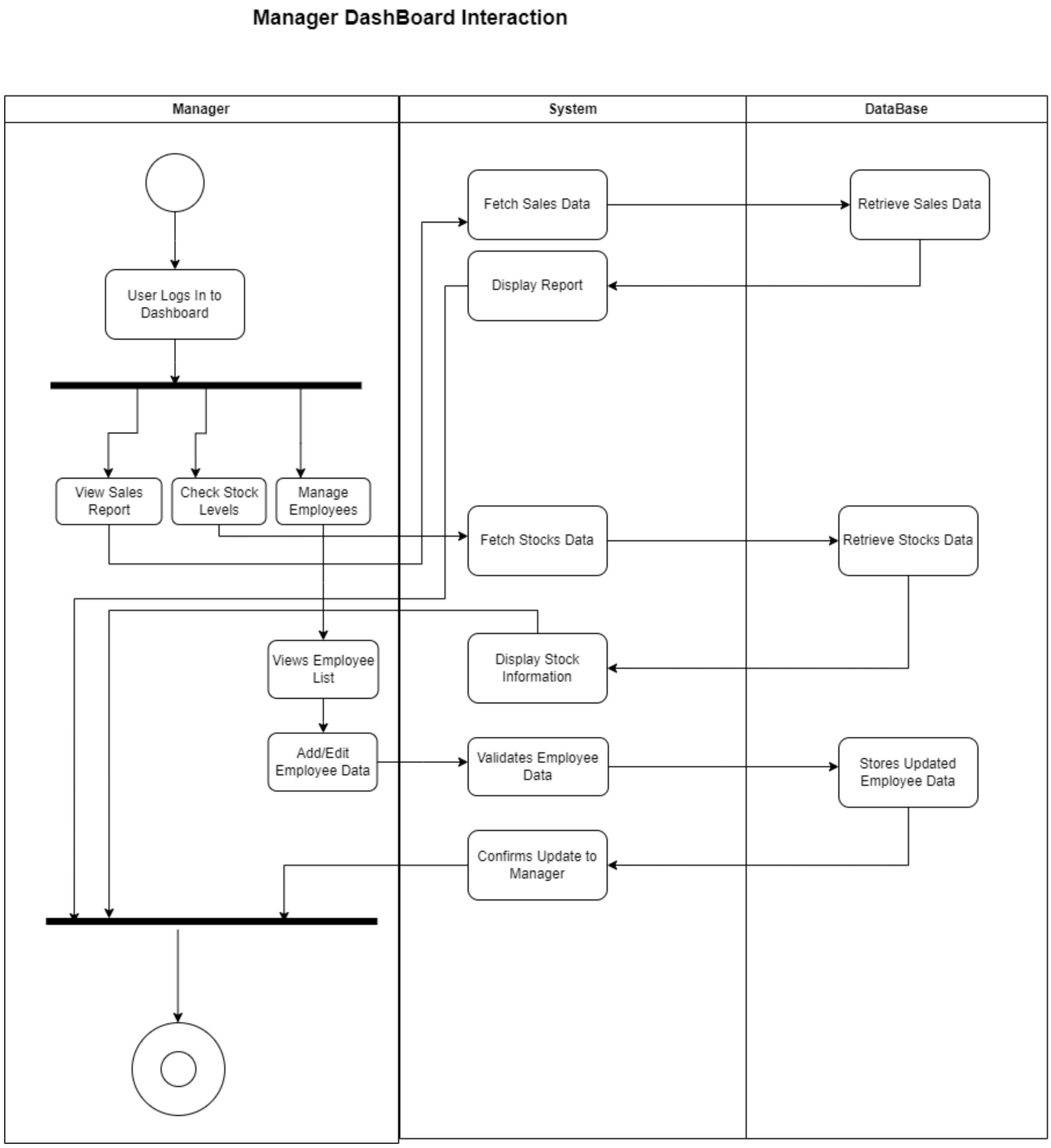

Figure 3.8: Manager Dashboard Interaction

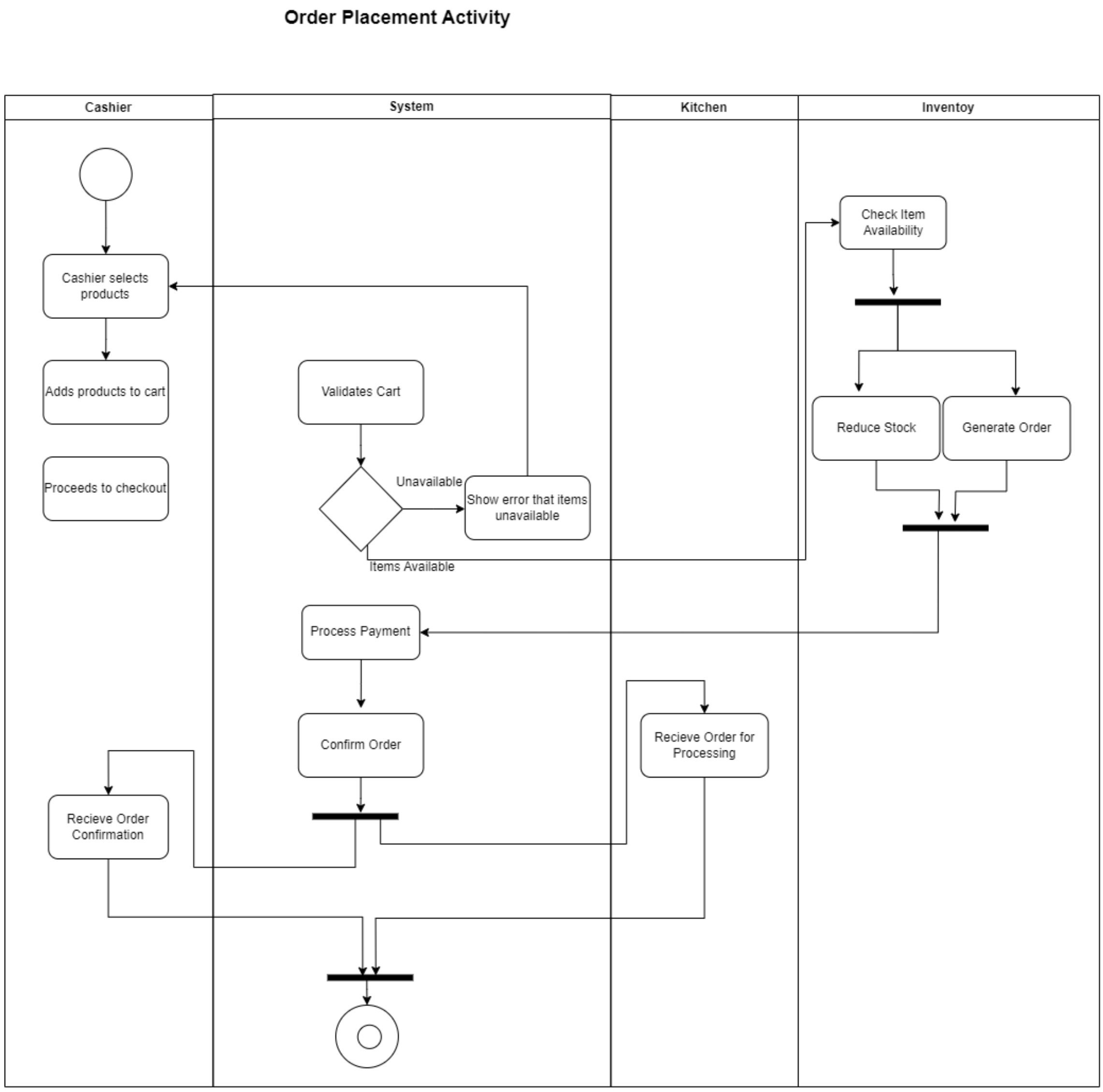

Figure 3.9: Order Placement

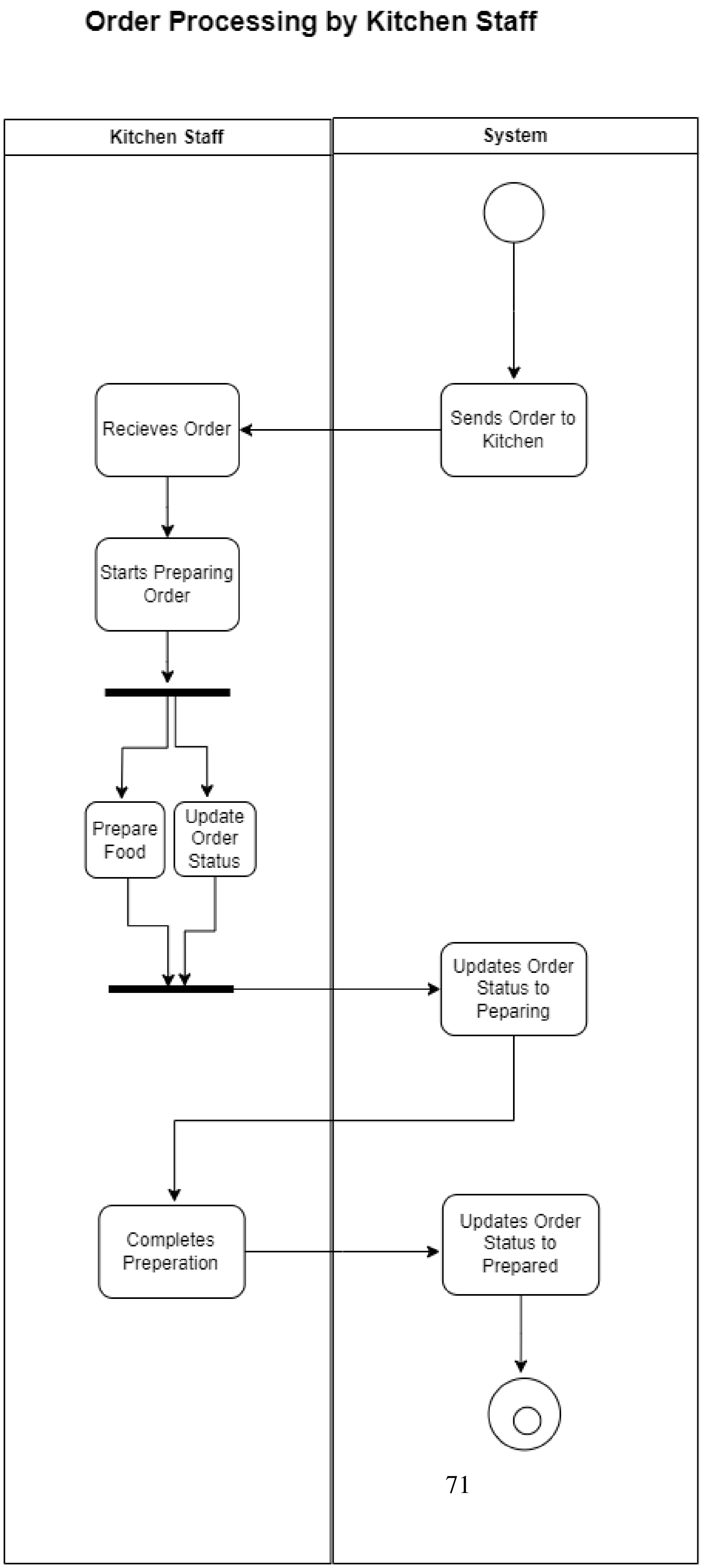

Figure 3.10: Order Processing

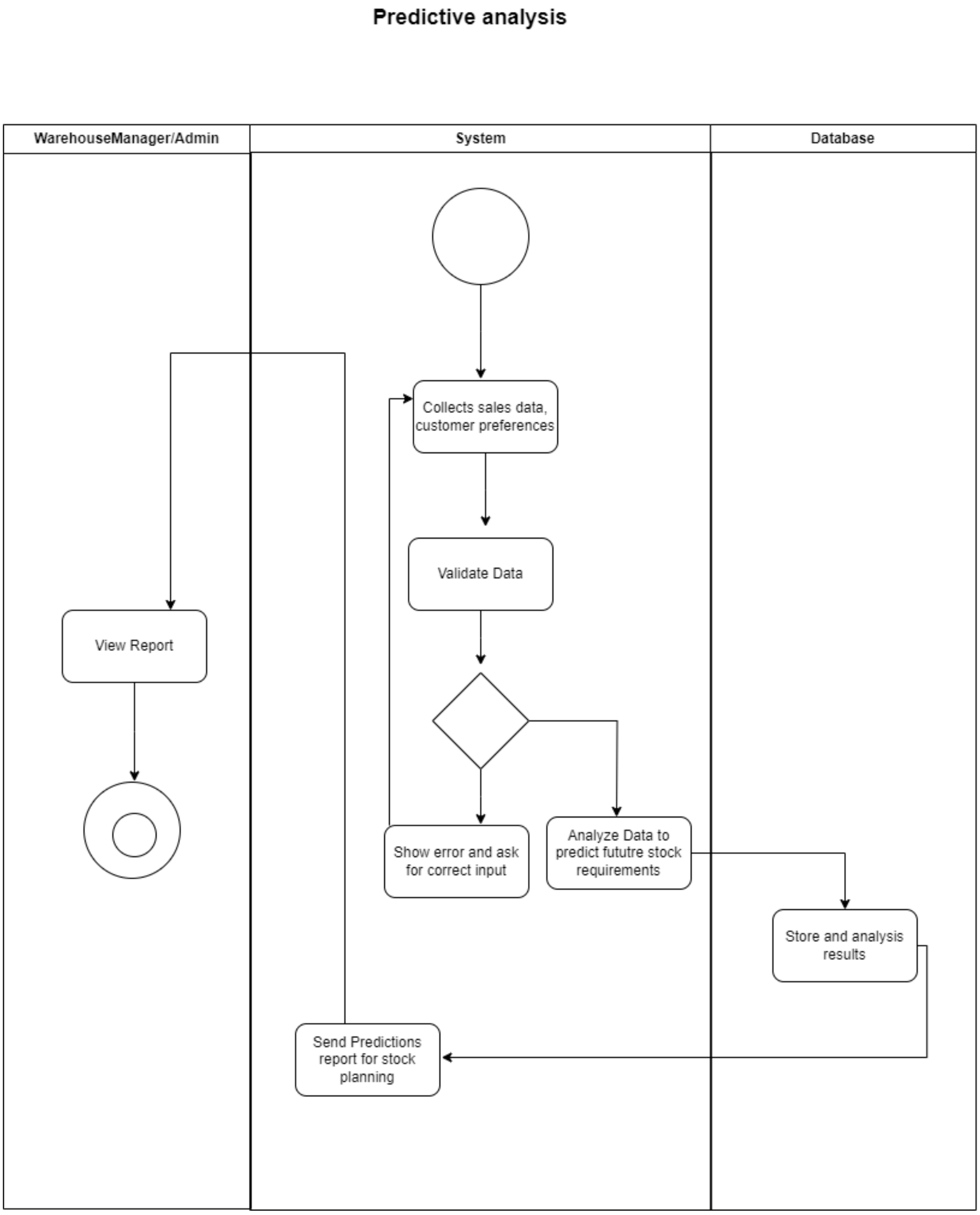

Figure 3.11: Predictive analysis

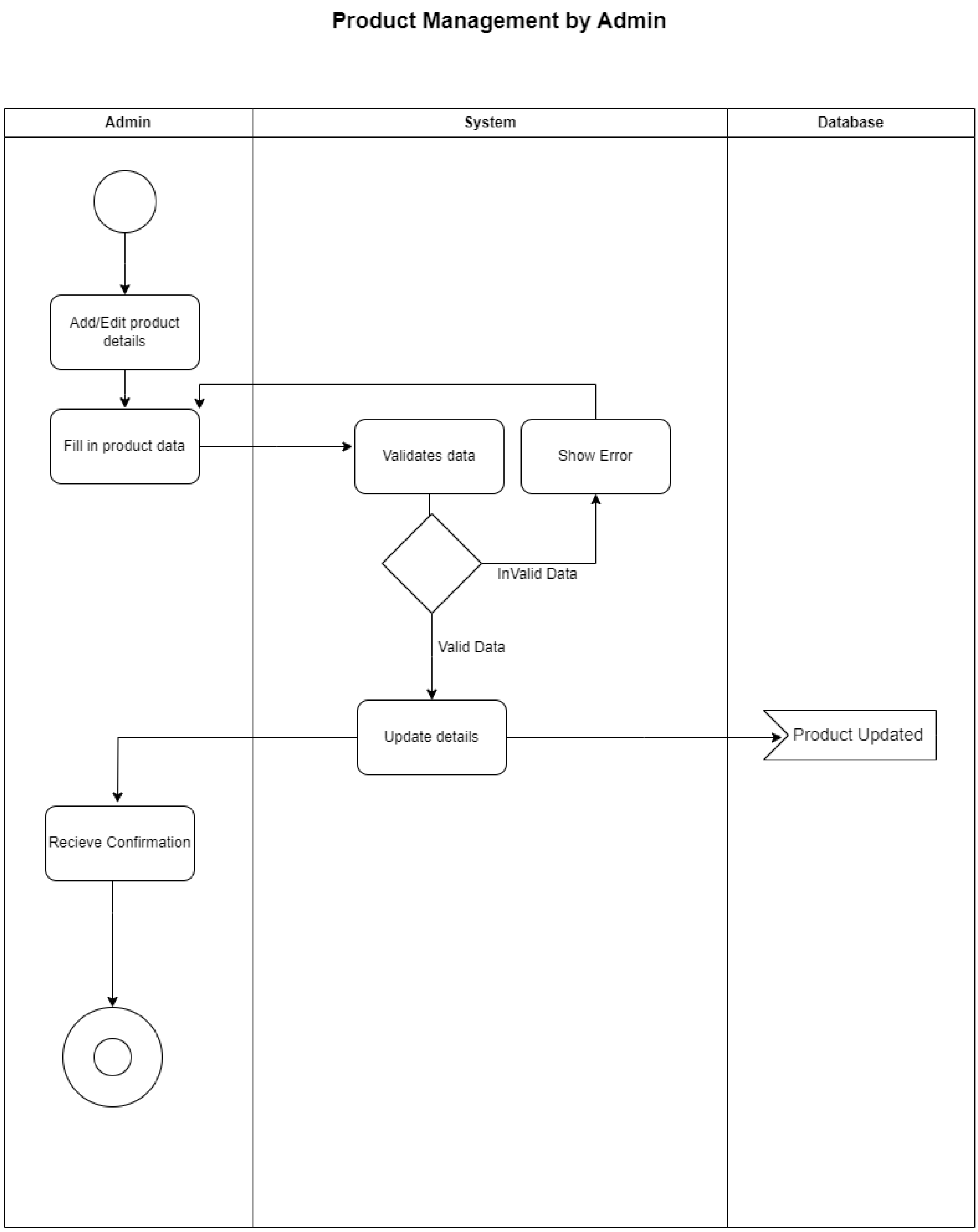

Figure 3.12: Product Management By Admin

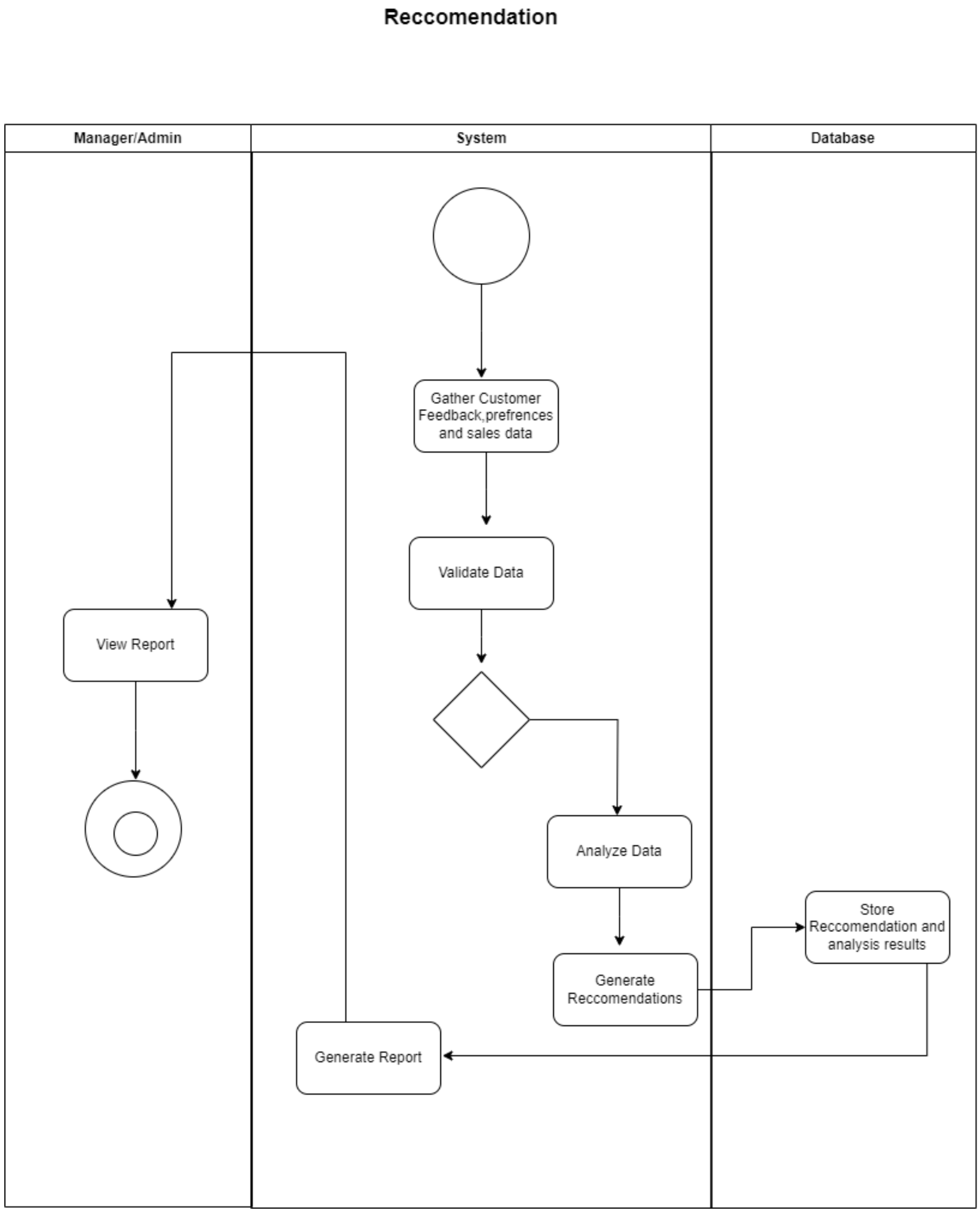

Figure 3.13: Recommendation

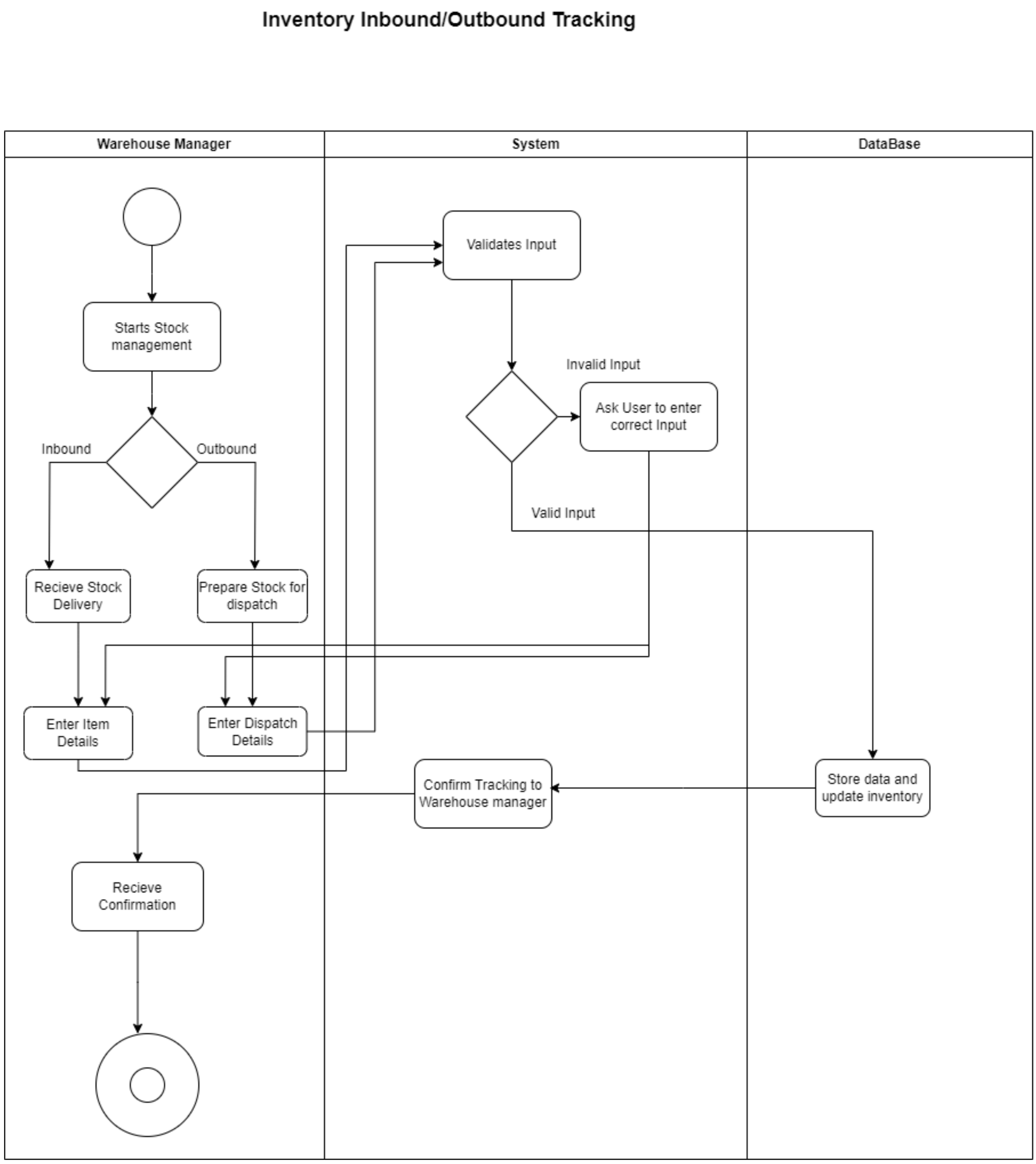

Figure 3.14: Inventory Inbound/Outbound Tracking

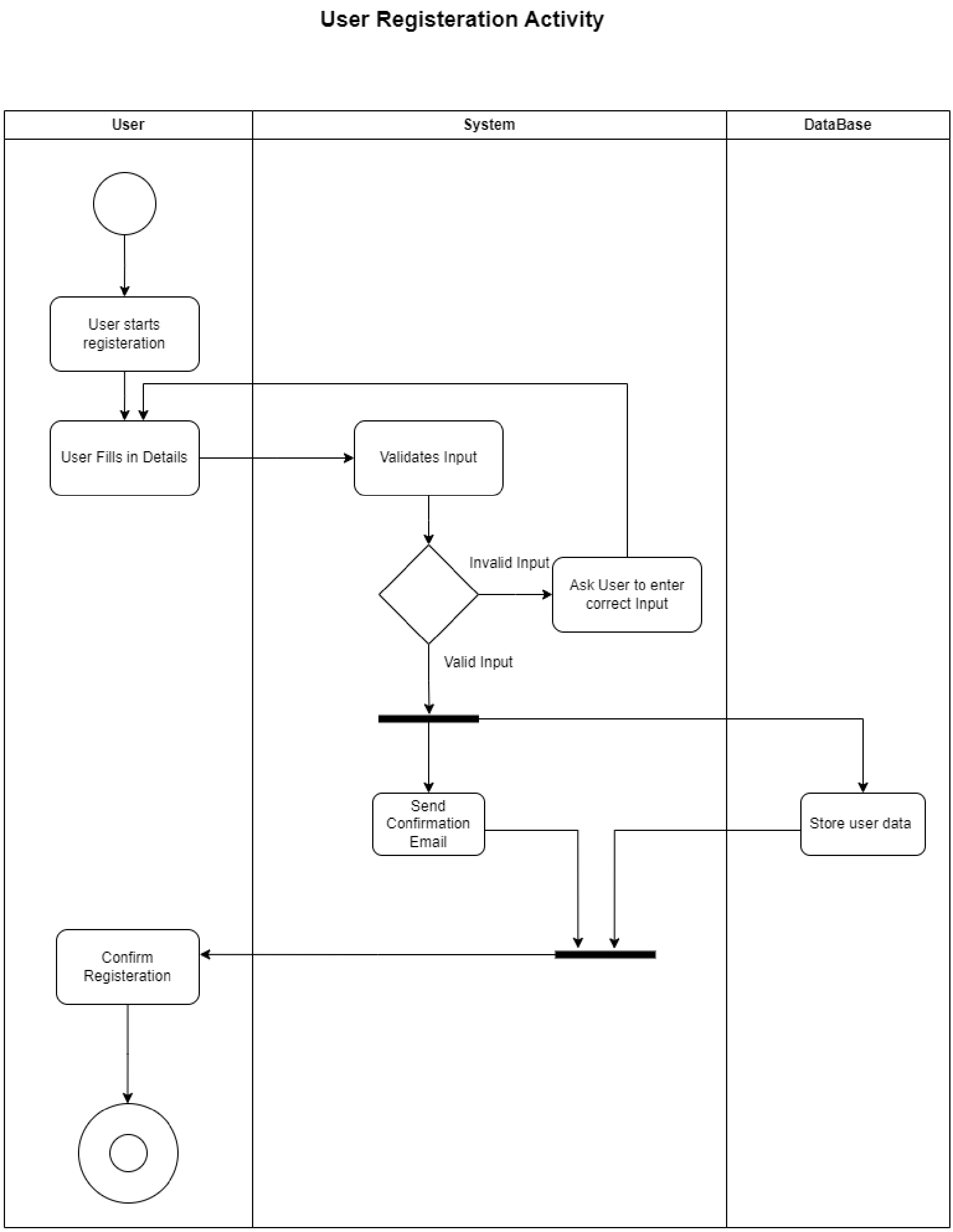

Figure 3.15: User Registration

3.1.2 Sequence Diagrams

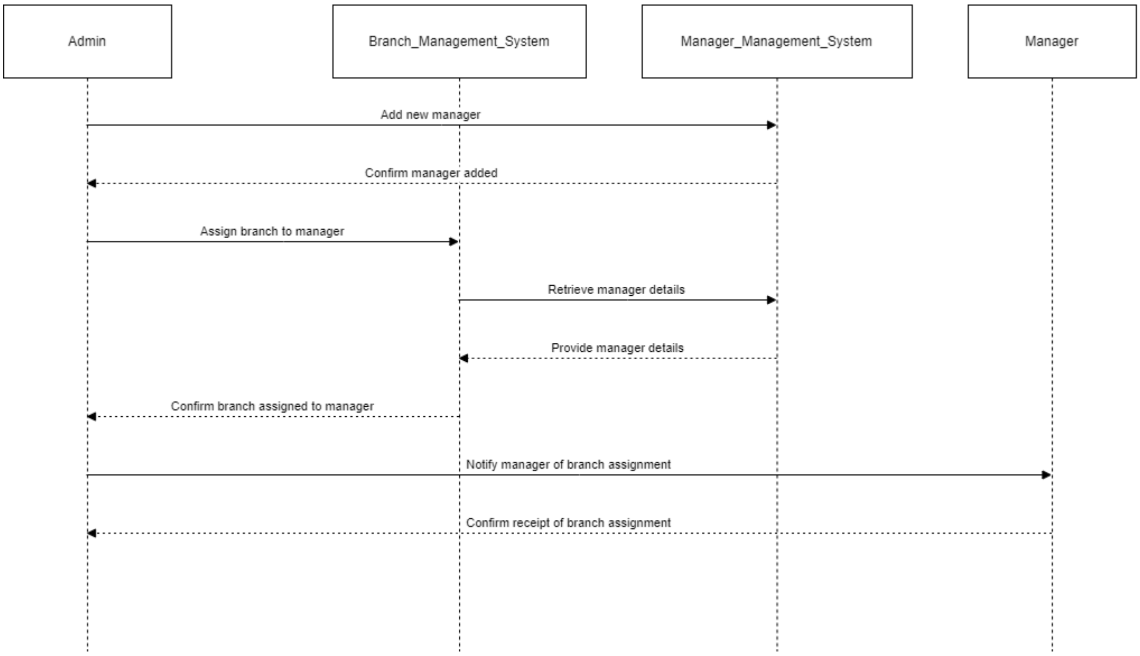

Figure 3.16: Branch Management

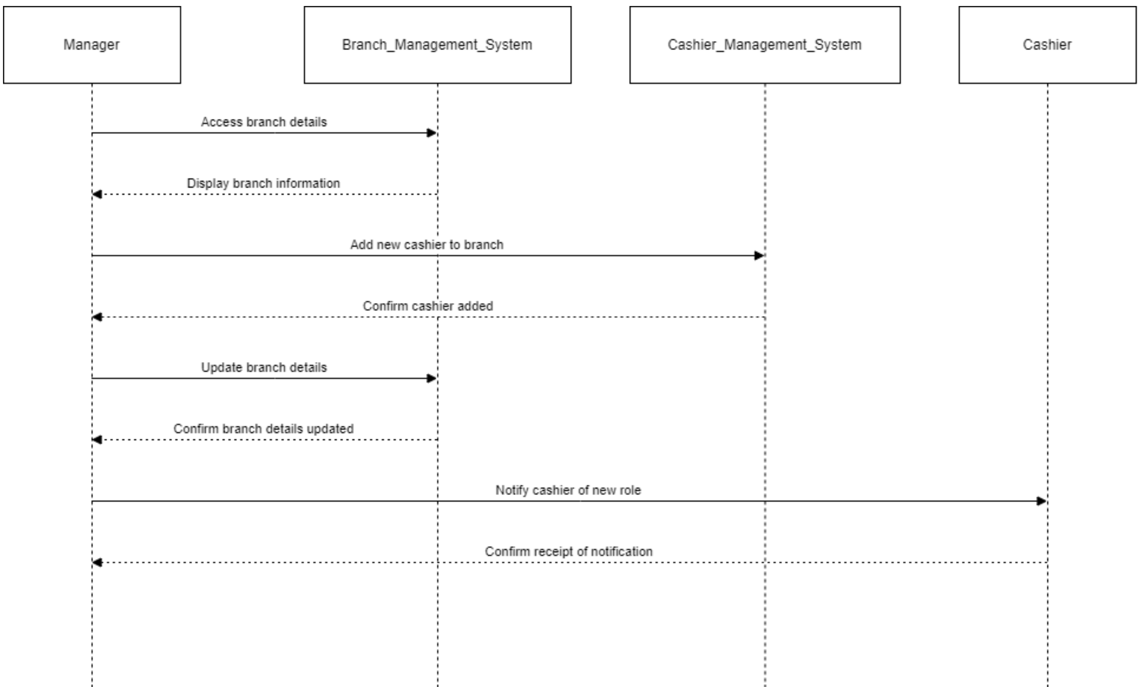

Figure 3.17: Cashier Management

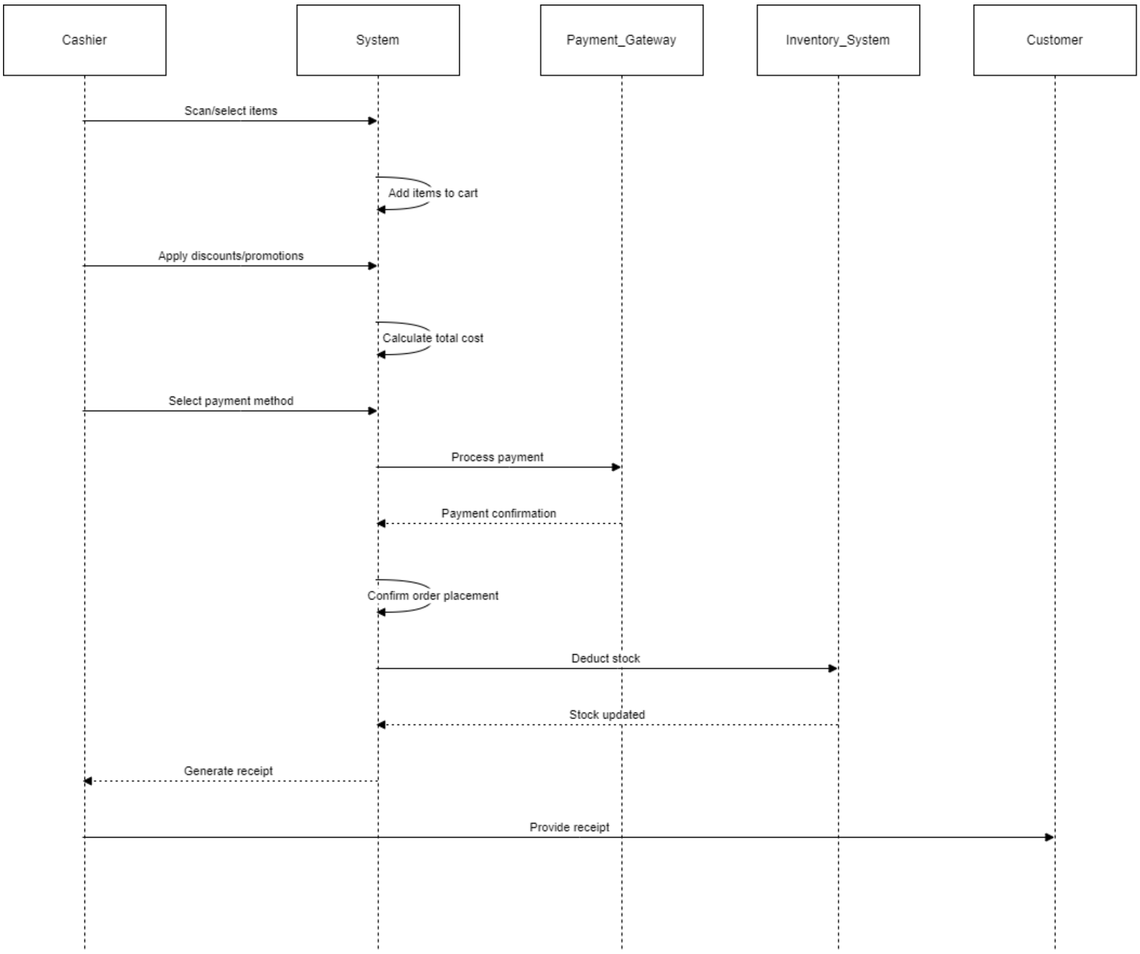

Figure 3.18: Cashier Order Processing

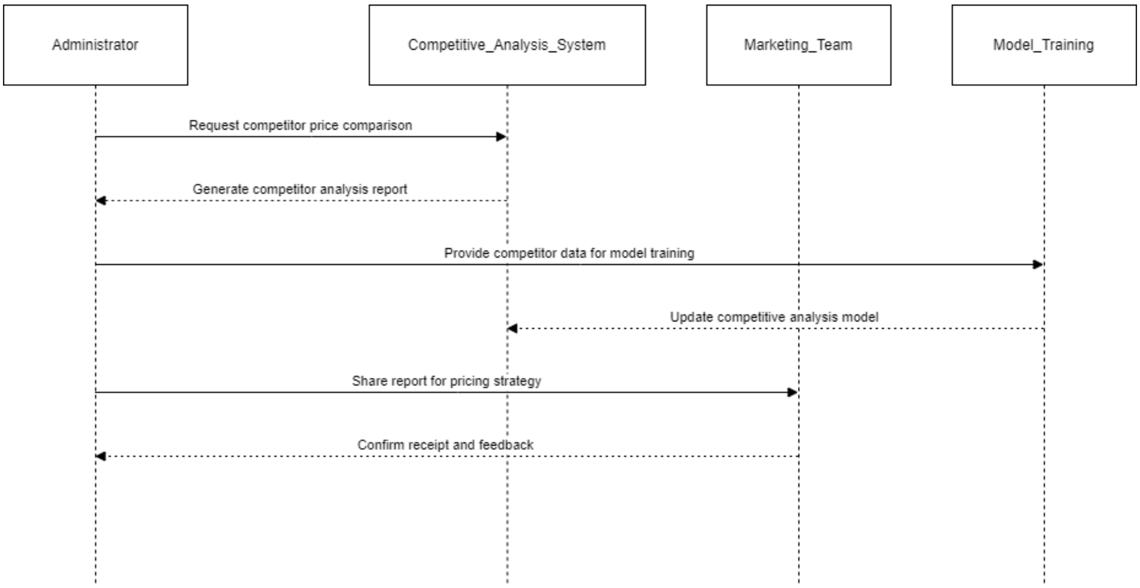

Figure 3.19: Competitive Analysis

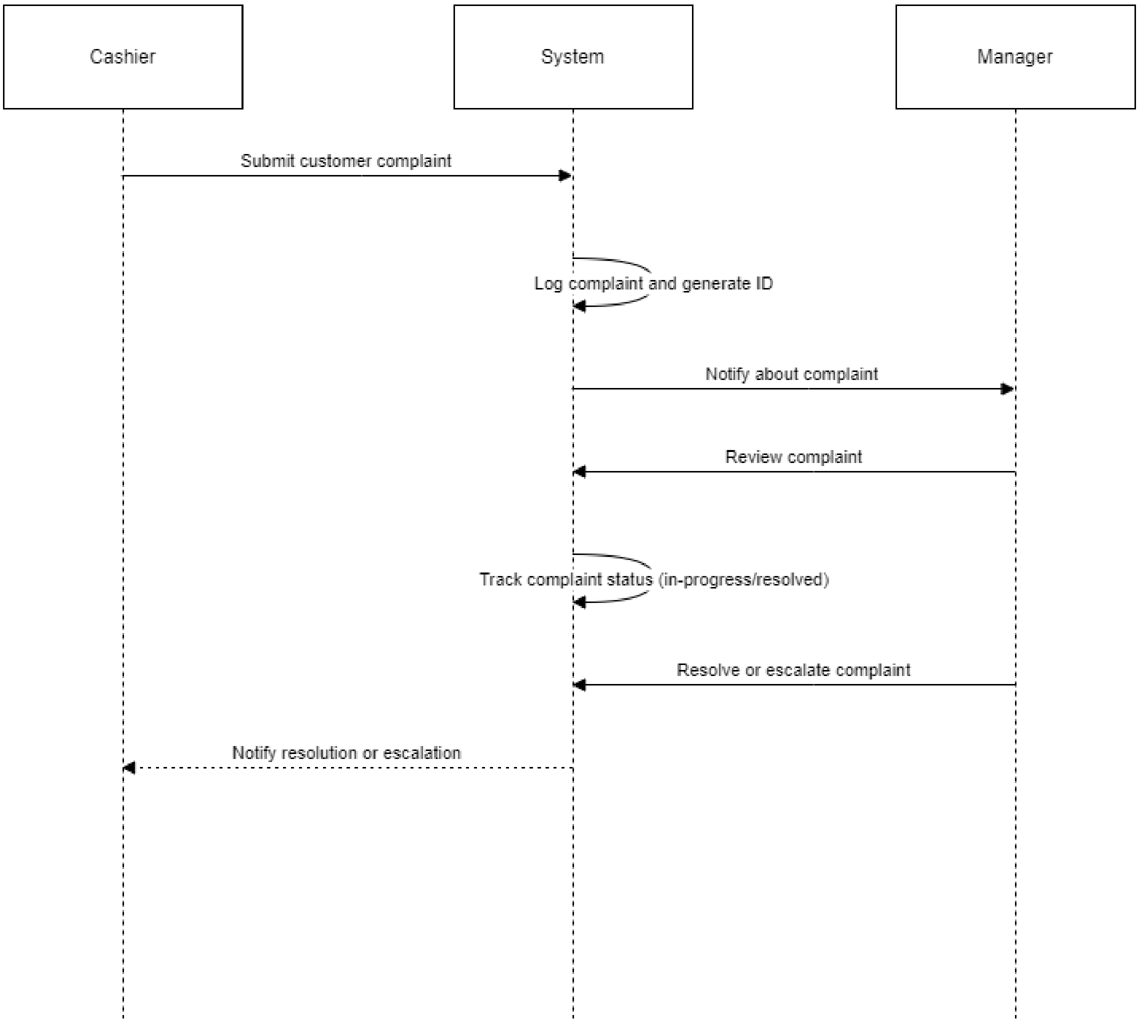

Figure 3.20: complaint Logging

Figure 3.21: Customer Feedback

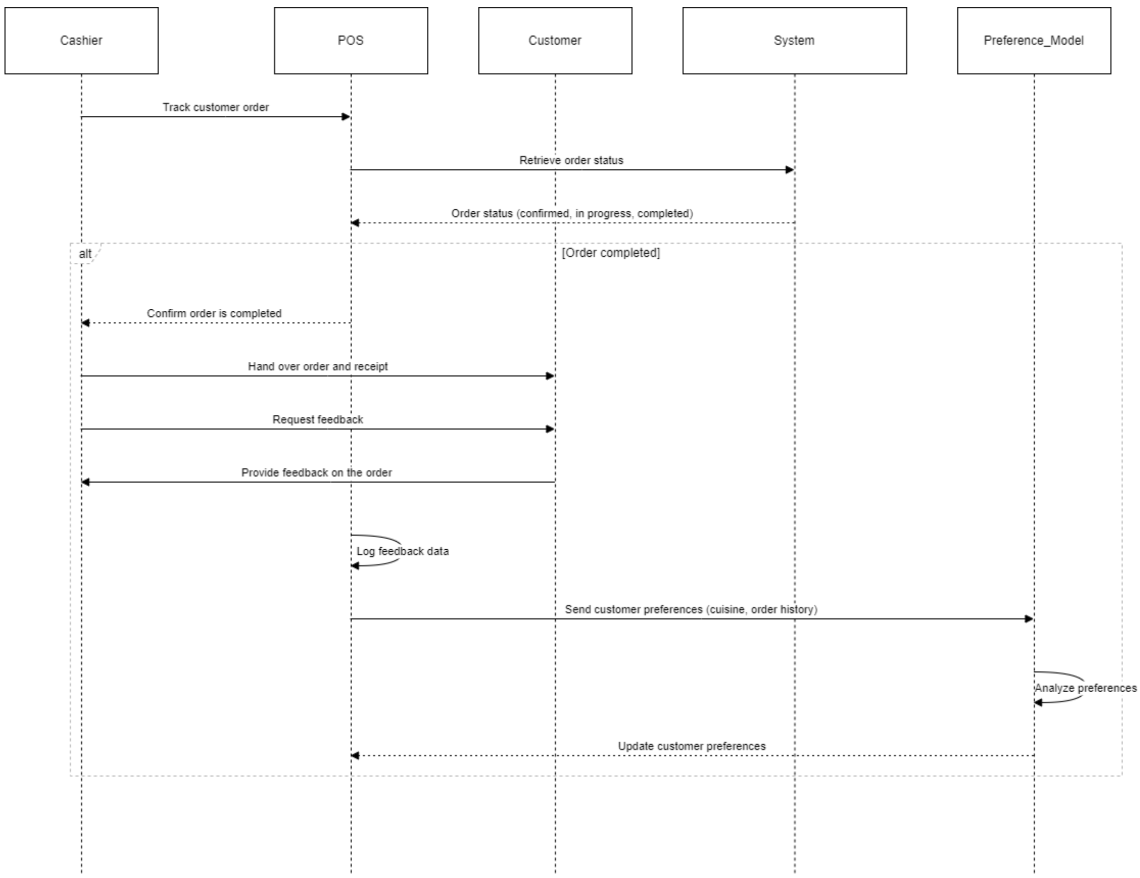

Figure 3.22: Customer Preferences Model

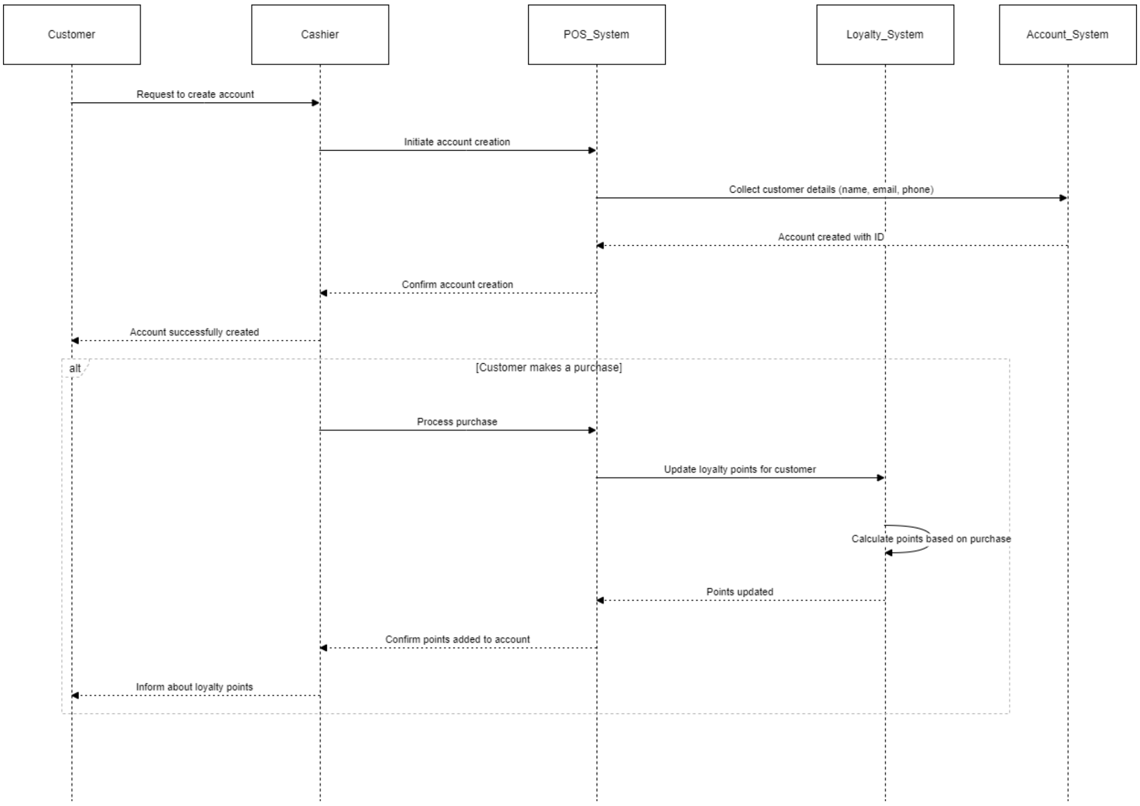

Figure 3.23: Customer Profile

Figure 3.24: Kitchen Module

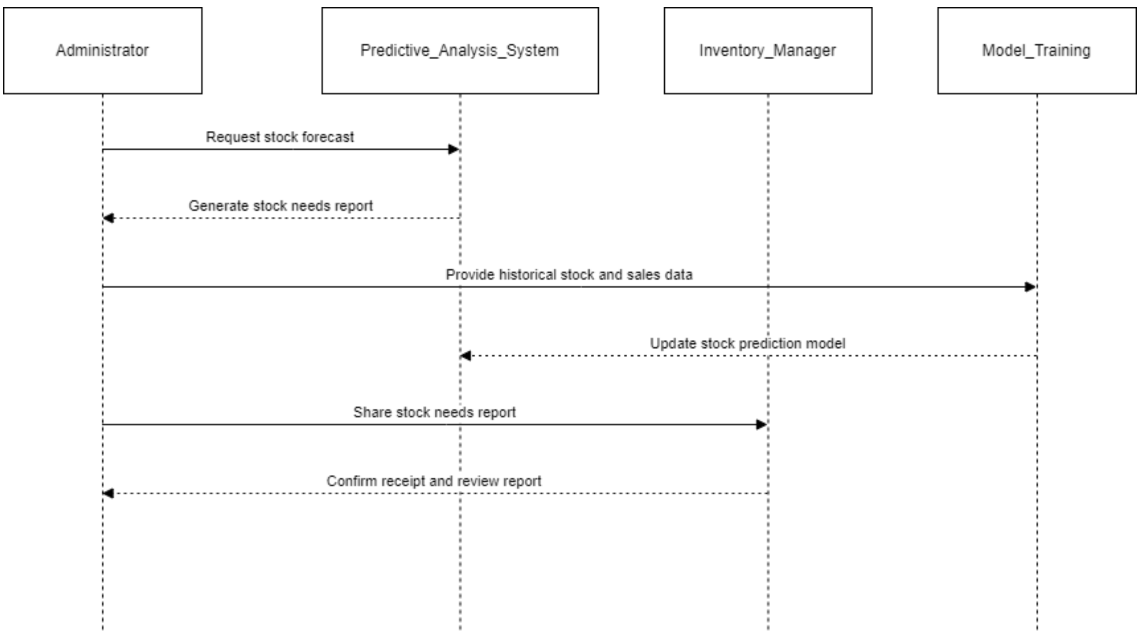

Figure 3.25: Predictive Stock Forecast

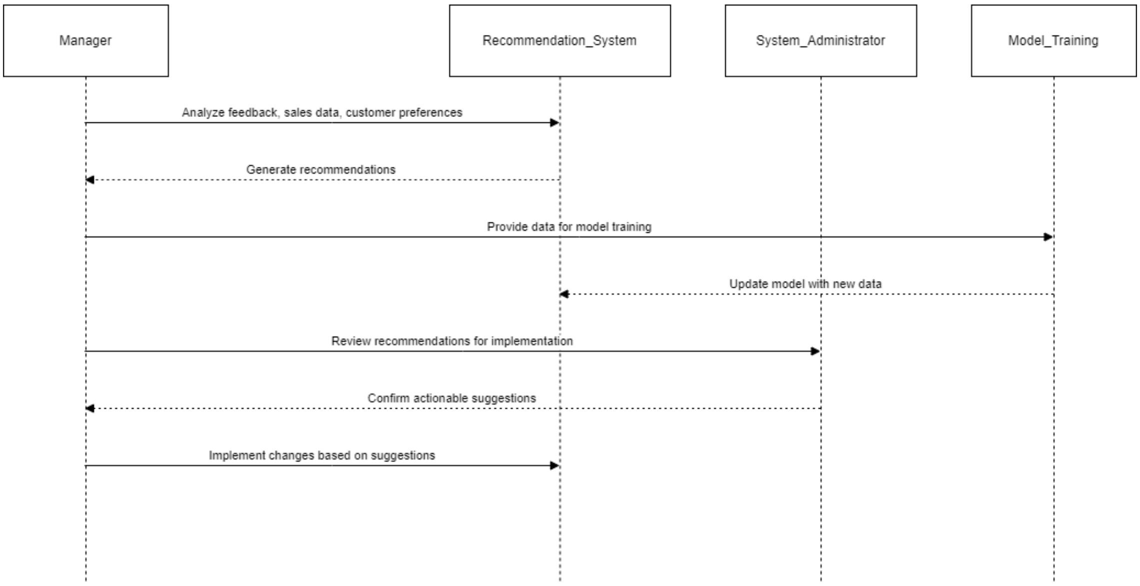

Figure 3.26: Recommendation System

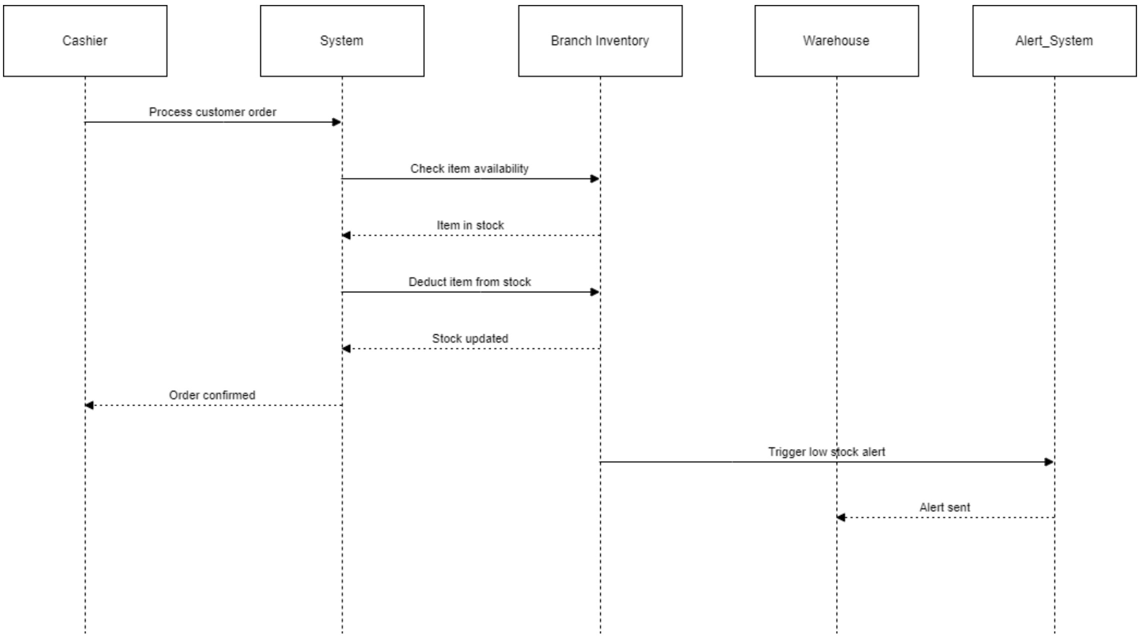

Figure 3.27: Stock Alert

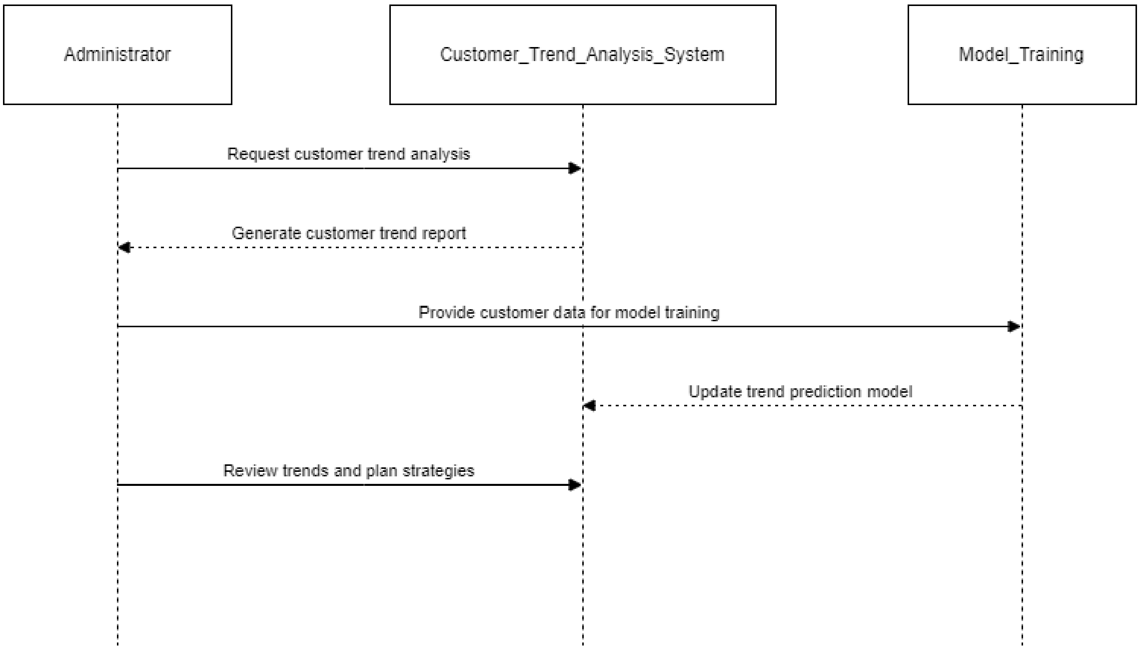

Figure 3.28: Trend Prediction

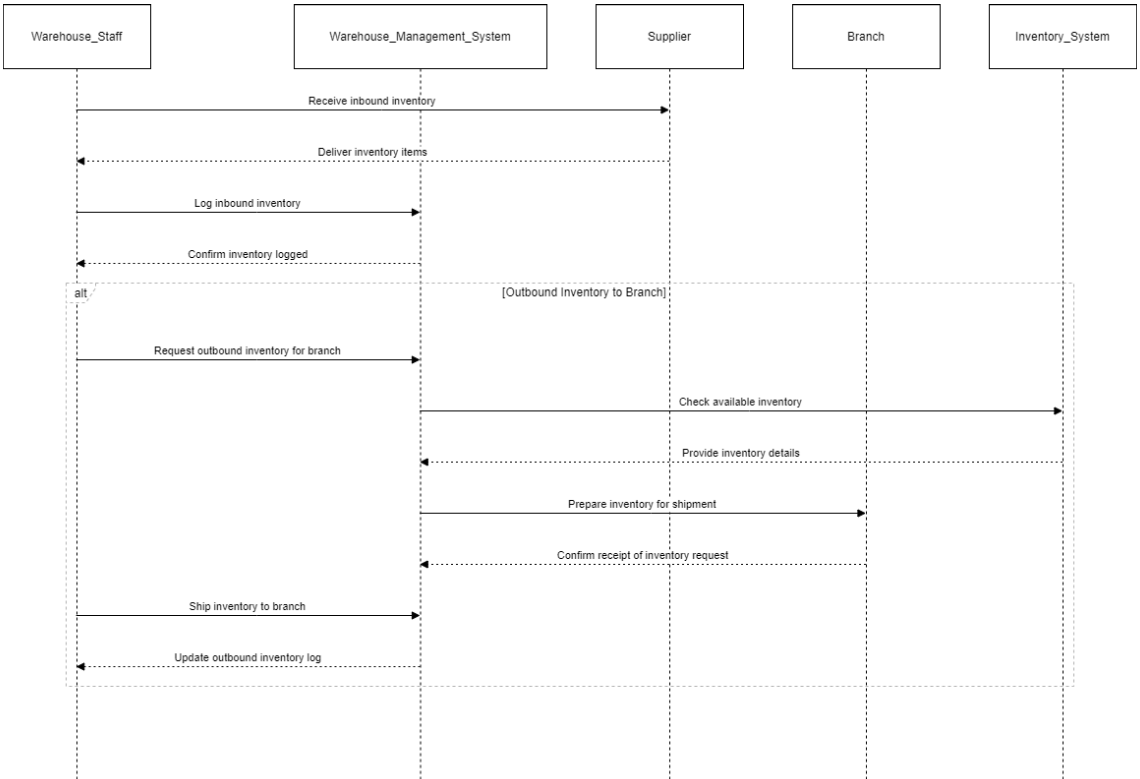

Figure 3.29: Warehouse

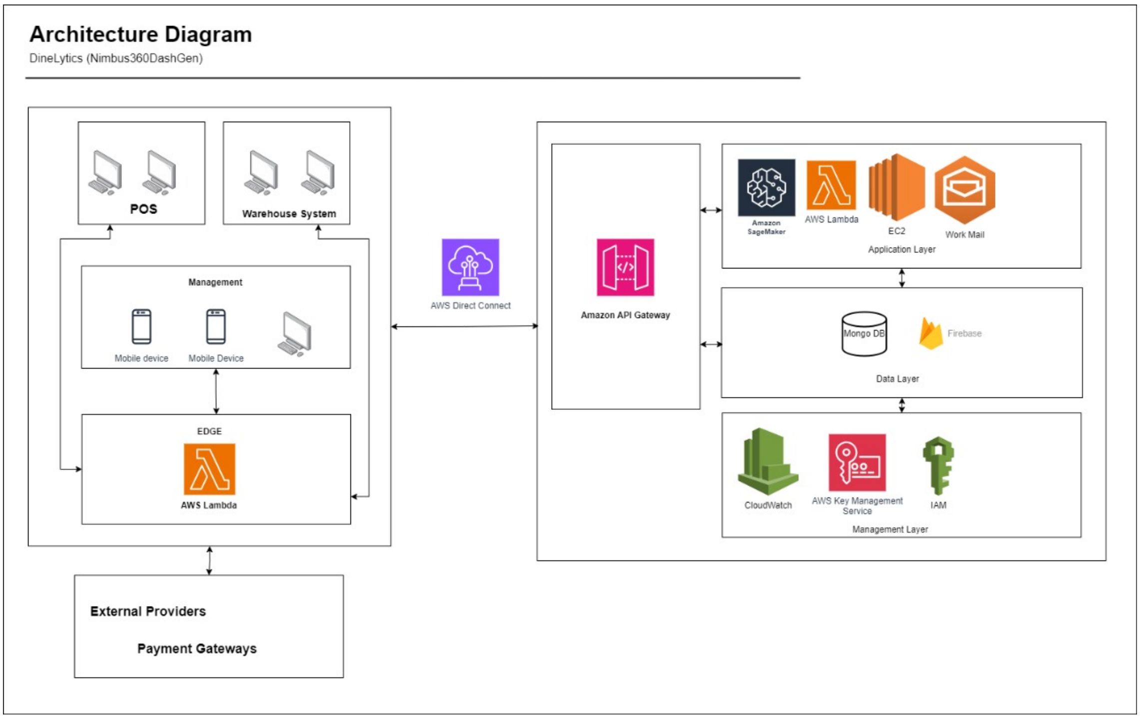

Figure 3.30: Architect Diagram

3.1.3 Architecture Diagram

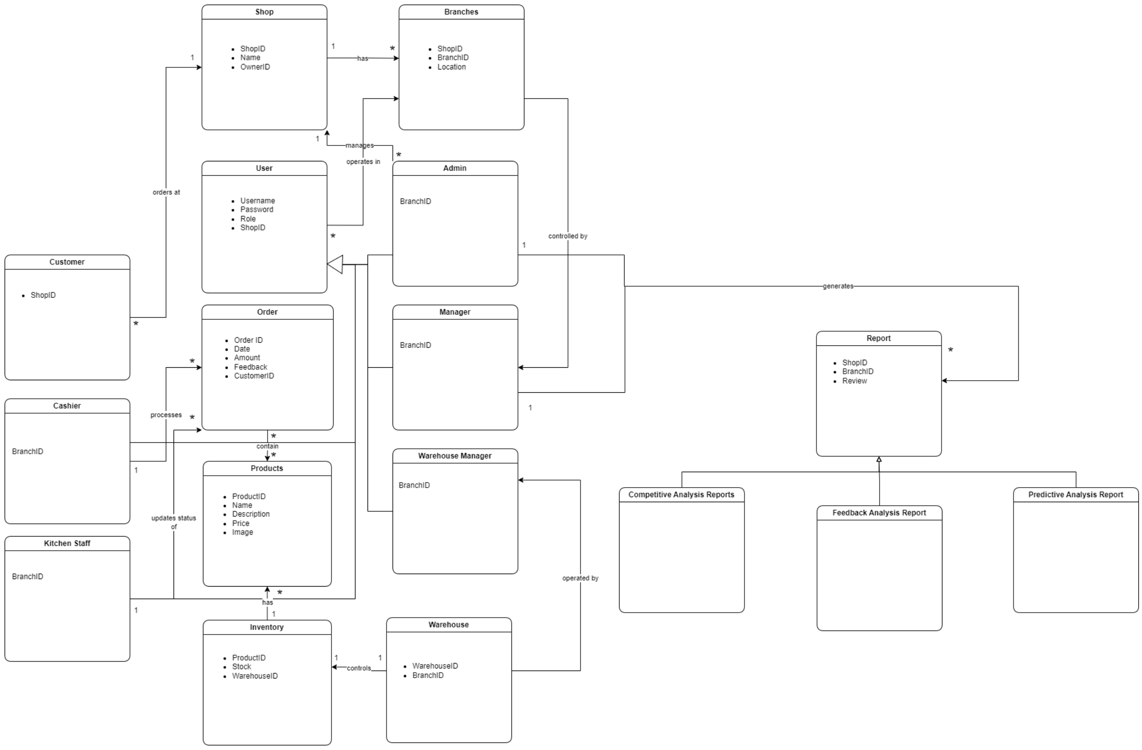

Figure 3.31: Domain Model

3.1.4 Domain Model

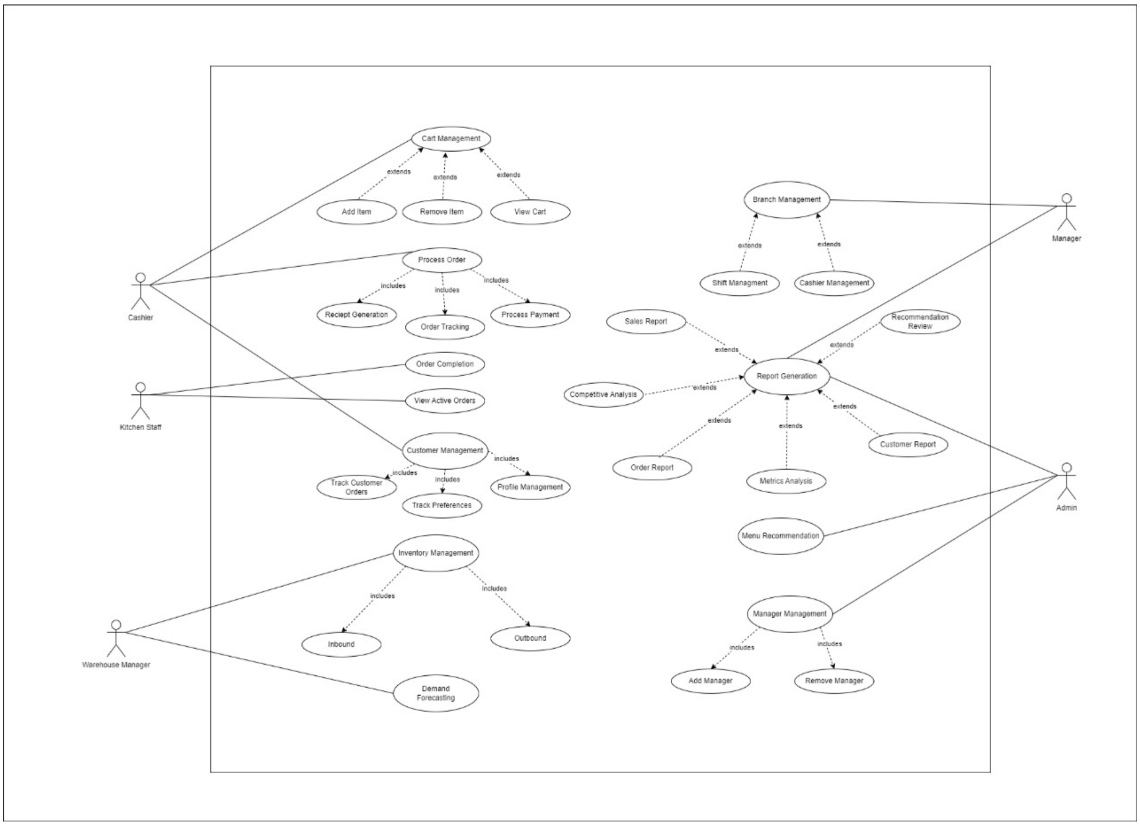

Figure 3.32: Use Case Diagram

3.1.5 Use Case Diagram

3.2 Data Design

3.2.1 Data Flow Diagram

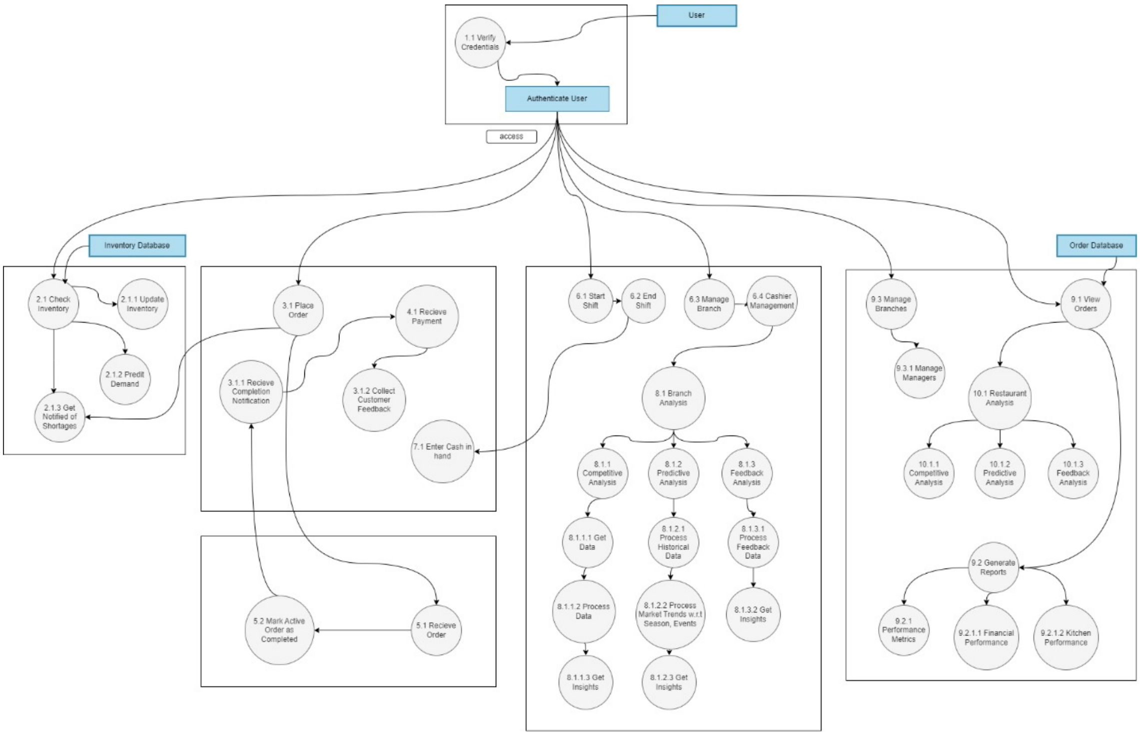

Figure 3.33: Data Flow Diagram

3.2.2 JSON Schema

User Collection

"type": "object",
"required": ["userID", "email", "password", "role"],
"properties":
"userID": "type": "string" ,
"email": "type": "string", "format": "email" ,
"password": "type": "string" ,
"name": "type": "string" ,
"role": "type": "string", "enum": ["Admin",
"Manager", "Employee", "Customer"] ,
"profile":
"type": "object",
"properties":
"address": "type": "string" ,
"phoneNumber": "type": "string"

,
"registeredAt": "type": "string", "format": "date-time" ,
"updatedAt": "type": "string", "format": "date-time"

Order Collection:

"type": "object",
"required": ["orderID", "customerID", "orderDate", "status", "totalAmount", "products"],
"properties":
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"customerID": "type": "string" ,
"orderDate": "type": "string", "format": "date-time" ,
"status": "type": "string", "enum": ["pending", "completed", "cancelled"] ,
"totalAmount": "type": "number" ,
"products":
"type": "array",
"items":
"type": "object",
"properties":
"productID": "type": "string" ,

"quantity": "type": "integer" ,
"price": "type": "number"

,
"payment":
"type": "object",
"properties":
"paymentID": "type": "string" ,
"paymentDate": "type": "string", "format": "date-time" ,
"amount": "type": "number"

Inventory Collection:

"type": "object",
"required": ["itemID", "name", "quantity"],
"properties":
"itemID": "type": "string" ,
"name": "type": "string" ,
"quantity": "type": "integer" ,
"lastUpdated": "type": "string", "format": "date-time"

Competitive Analysis Collection:

"type": "object",
"required": ["analysisID", "generatedAt"],
"properties":
"analysisID": "type": "string" ,
"generatedReport": "type": "string" ,
"generatedAt": "type": "string", "format": "date-time"
,
"Recommendations": "type": "string" ,

Scraped Data:

"type": "object",

"required": ["scrapeID", "dataSources", "scrapedAt"],
"properties":
"scrapeID": "type": "string" ,
"dataSources":
"type": "array",
"items": "type": "string"
,
"scrapedData": "type": "string" ,
"scrapedAt": "type": "string", "format": "date-time"

Payment Collection:

"type": "object",
"required": ["paymentID", "orderID", "paymentDate", "amount"],
"properties":
"paymentID": "type": "string" ,
"orderID": "type": "string" ,
"paymentDate": "type": "string", "format": "date-time" ,
"amount": "type": "number"

Product Collection:

"type": "object",
"required": ["productID", "name", "price", "description", "imageURL", "available"],
"properties":
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"name": "type": "string" ,
"description": "type": "string" ,
"price": "type": "number" ,
"imageURL": "type": "string", "format": "uri" ,
"available": "type": "boolean" ,
"createdAt": "type": "string", "format": "date-time" ,
"updatedAt": "type": "string", "format": "date-time"

Restaurant Collection:

"type": "object",
"required": ["restaurantID", "name", "address", "branches", "rating"],
"properties":
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"name": "type": "string" ,
"address": "type": "string" ,
"branches":
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"items": "type": "string"
,
"rating":
"type": "object",
"properties":
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"reviewCount": "type": "integer"

,
"createdAt": "type": "string", "format": "date-time" ,
"updatedAt": "type": "string", "format": "date-time"

Branch Collection:

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"properties":
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"restaurantID": "type": "string" ,
"address": "type": "string" ,
"managerID": "type": "string" ,
"staff":
"type": "array",
"items": "type": "string"
,
"inventory":
"type": "array",
"items": "type": "string"

,
"status": "type": "string", "enum": ["open", "closed", "under$_{m}aintenance”$],
$”createdAt” : ”type” : ”string”,”format” : ”date-time”$,
$”updatedAt” : ”type” : ”string”,”format” : ”date-time”$

Performance Metrics:

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Bibliography